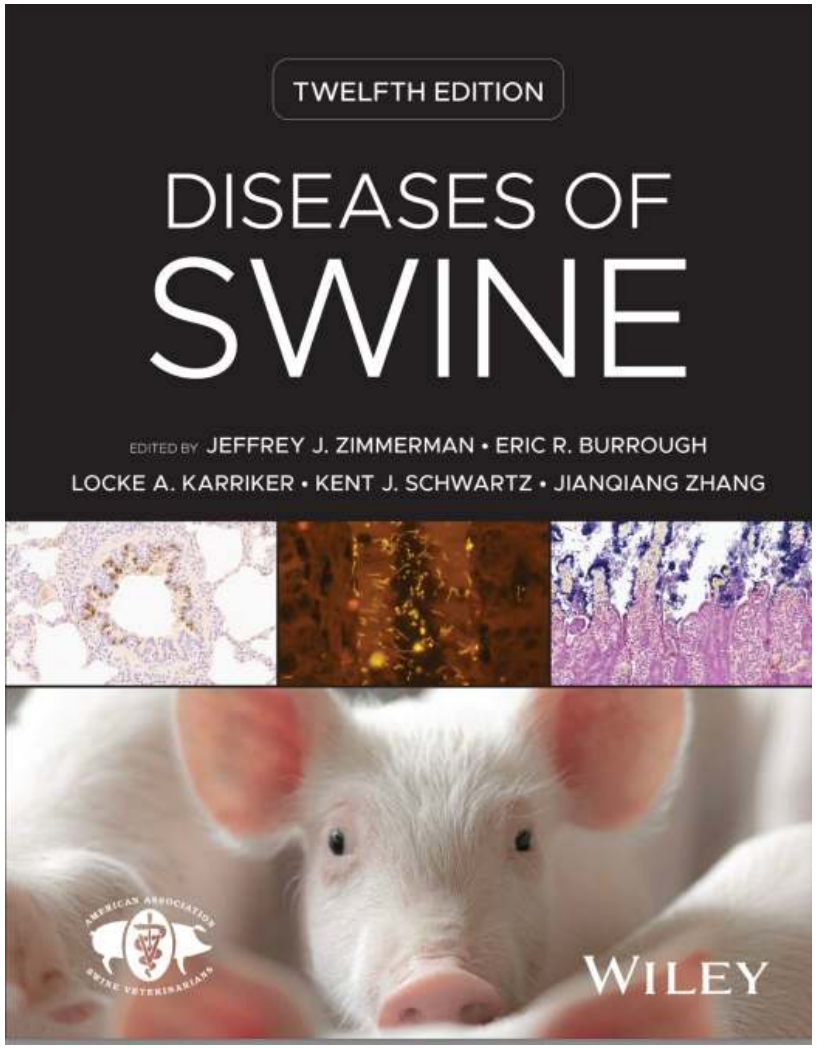
Swine viral diseases (I)

張家宜
國立臺灣大學 獸醫專業學院
2026.09.17





Textbook





Jeffrey Zimmerman

Iowa State University



Jianqiang Zhang

Iowa State University







9-12 November, 2025

Sustainable Future of Asian Pig Production

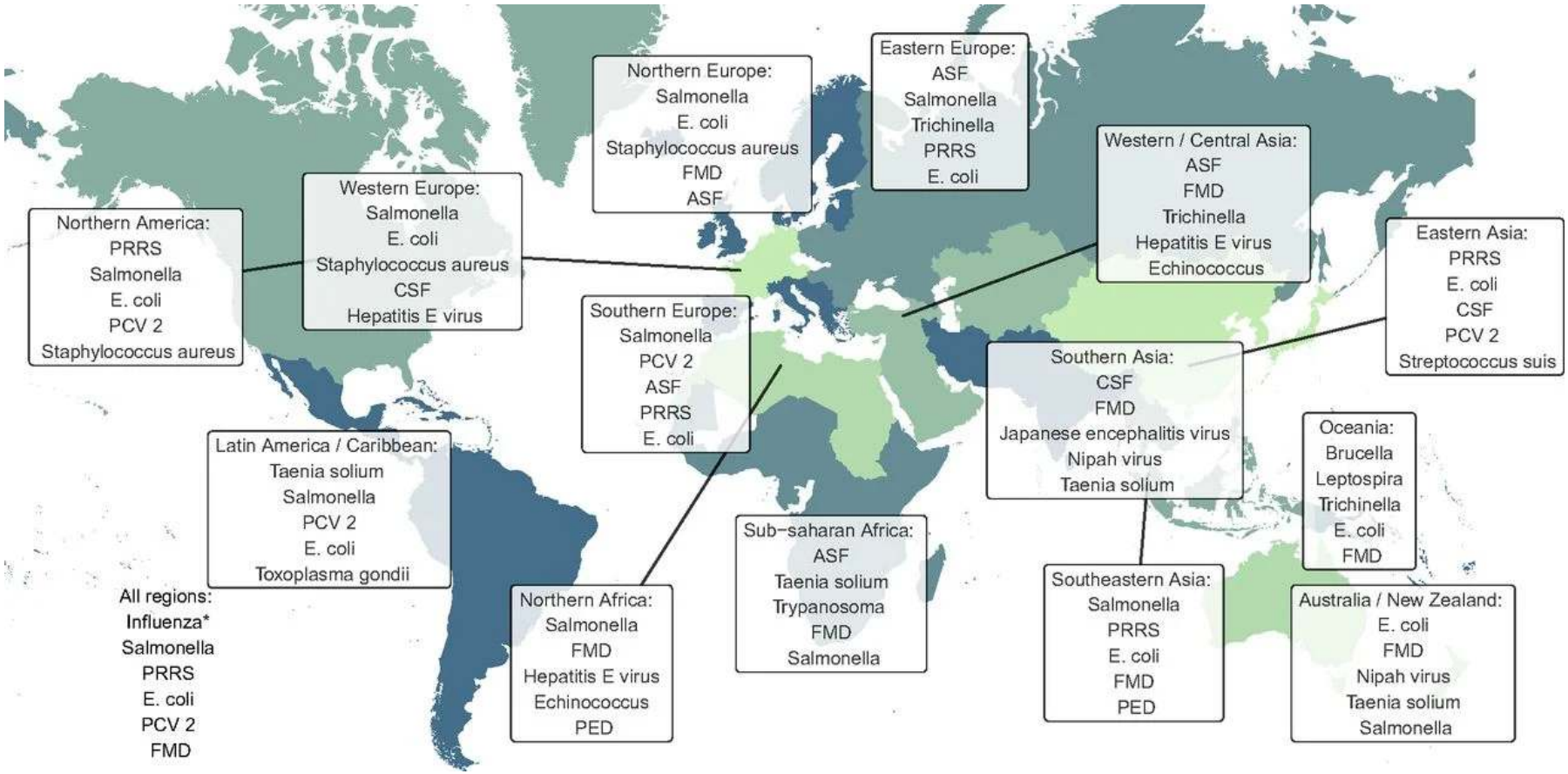




Outline

- African swine fever (ASF)
- Classical swine fever (CSF)
- Japanese encephalitis
- Hepatitis E

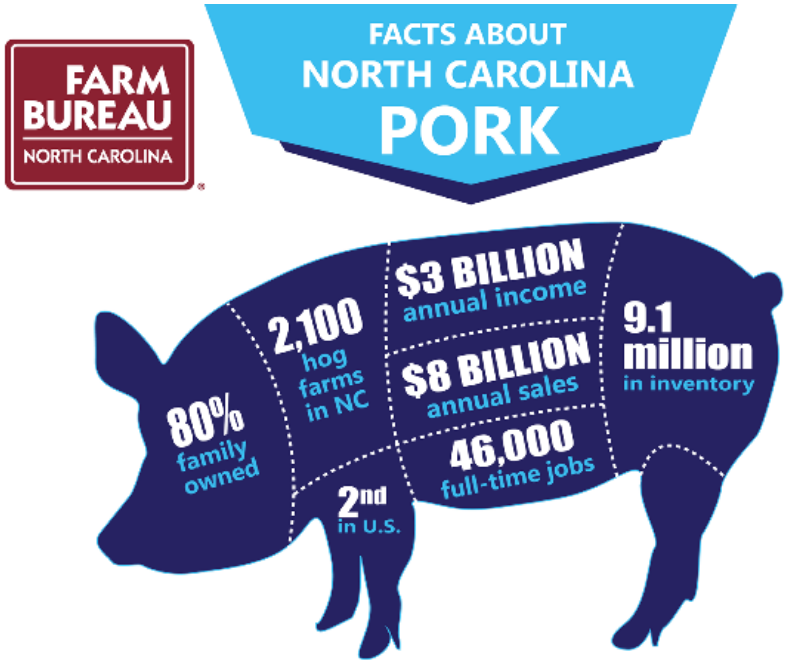
Most important pathogens of swine by region during 2006-2016



VanderWaal et al., 2018

The impact of swine diseases on the pig industry

- **Pigs** are a primary source of income in many countries.
- The global spread of major swine diseases can have devastating impacts on the pig industry.
- Biodiversity and ecosystem balance: these diseases affect not only domestic farmed pigs but also **wild boars**.



Impact of swine diseases on food security and food safety

- **Pork** meat is one of the primary sources of animal proteins, accounting for more than **35%** of the global meat intake.



- Major swine diseases can threaten global **food security**, while zoonotic swine diseases also pose important **food-safety** risks.

Pig industry in China







African swine fever

RESEARCH HIGHLIGHT | 01 October 2021

The $100-billion toll of a pig epidemic in China

Researchers estimate that a 2018 outbreak of African swine fever virus led to the loss of at least 38 million more animals than officially acknowledged.

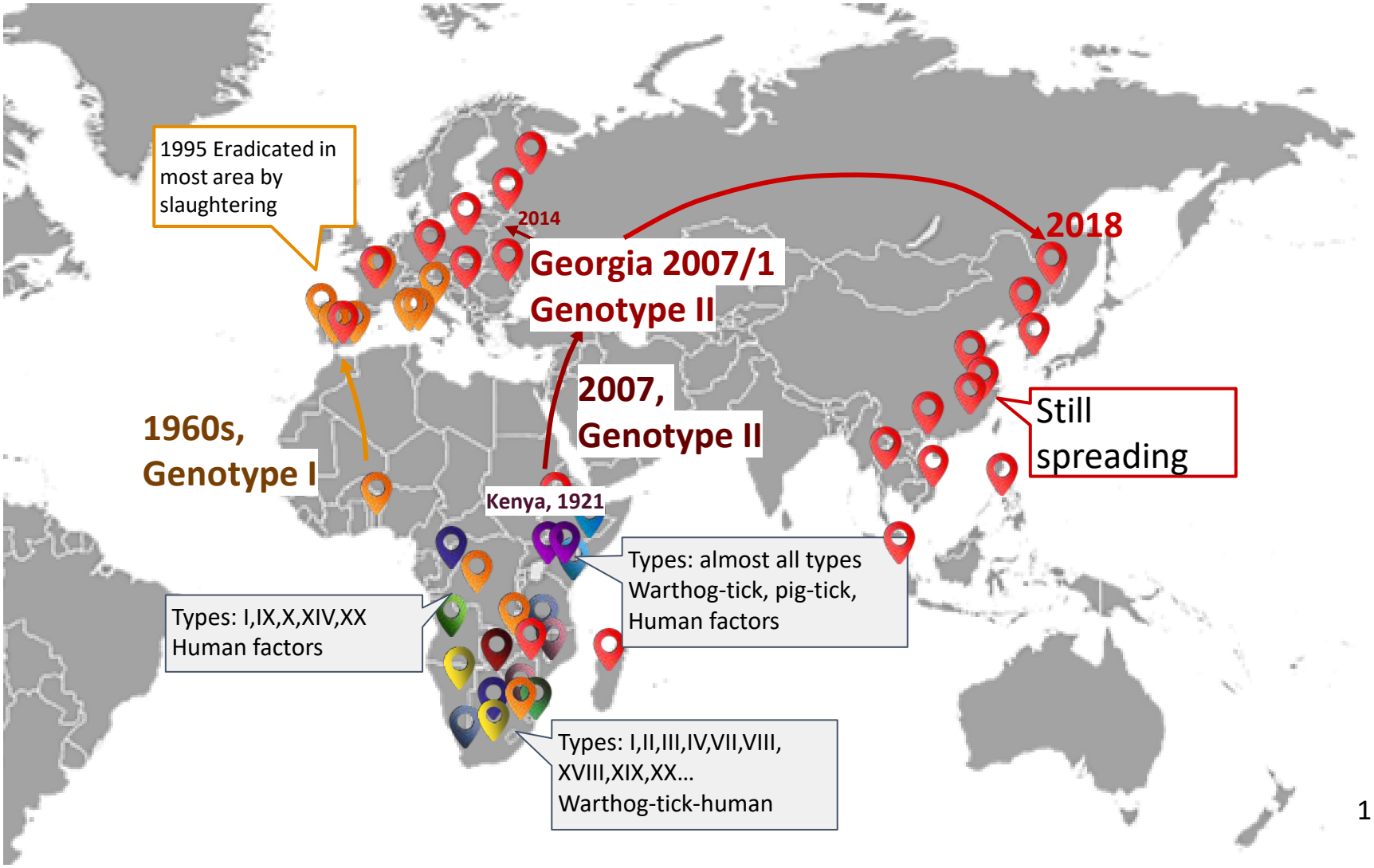


A farmer in China with some of the few pigs left in his herd after a cull.

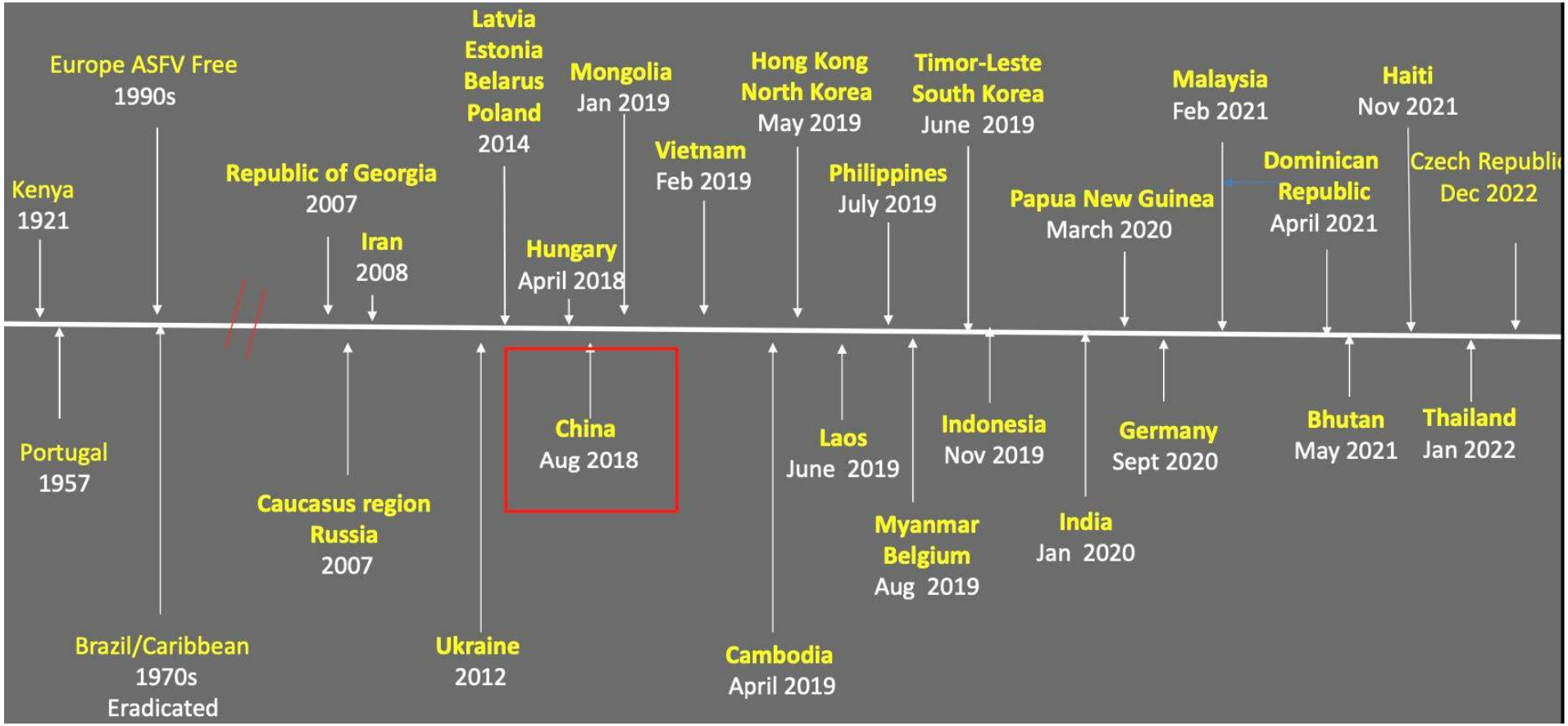
African swine fever

- African swine fever (ASF) is a highly contagious viral disease of **domestic pigs** and **wild boars**, with a mortality rate of up to 100%.
- ASF is **not** zoonotic but devastates pig populations and the swine economy.
- The virus is highly resistant in the environment and can survive for prolonged periods in certain pork products.
- Human activities can play an important role in the transboundary spread of ASF when appropriate control measures are not implemented.

Transcontinental spread of ASFV

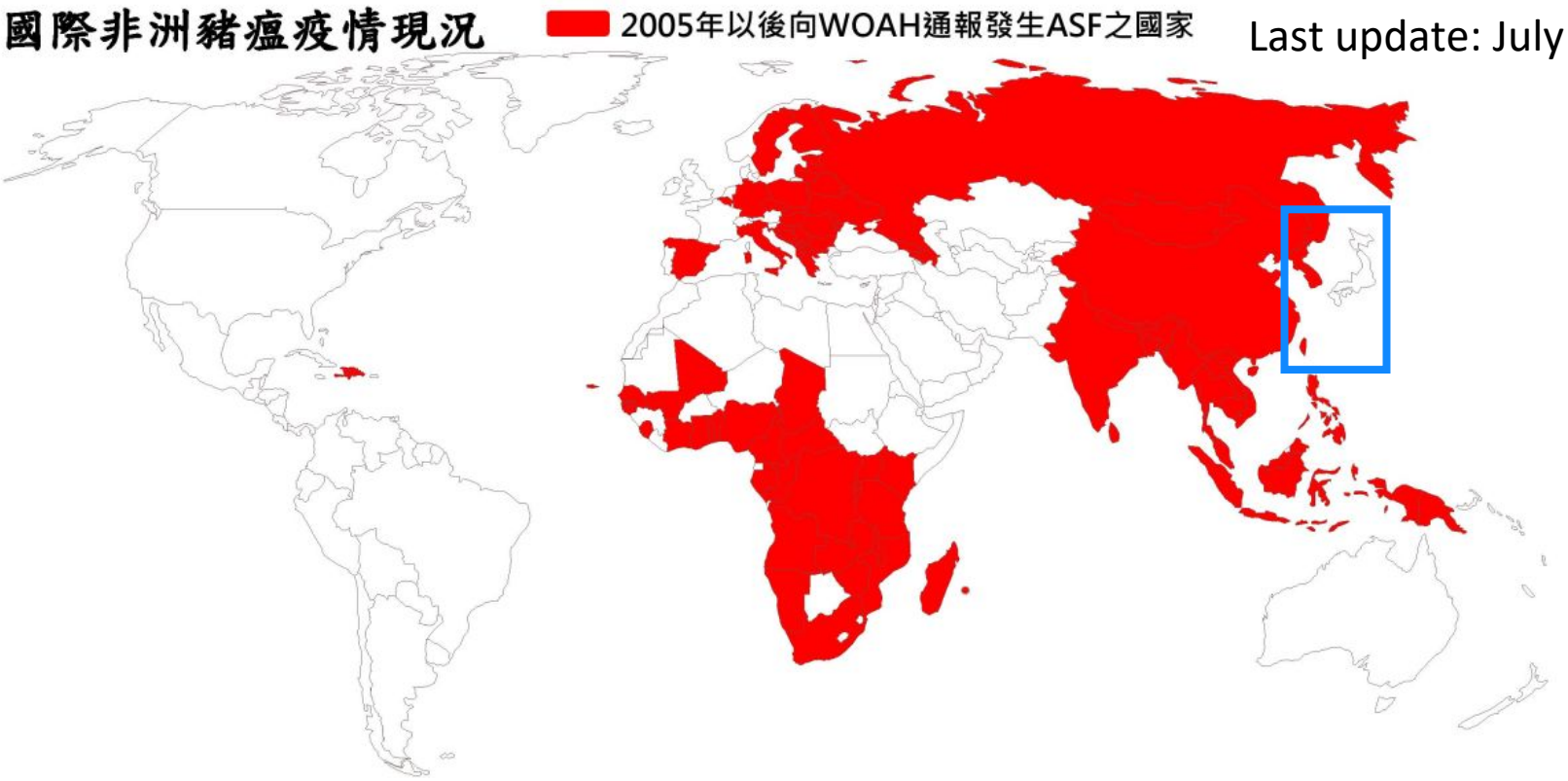


ASF outbreaks



Source: Dr. Gladue, USDA

Global situation of ASF 2026



Last update: July 31, 2026

非洲地區（計32國）

安哥拉 貝南 布吉納法索 蒲隆地 維德角 喀麥隆 中非 查德 民主剛果 剛果 獅子山 迦納 幾內亞比索 肯亞 馬達加斯加 馬拉威 馬利共和國 模里西斯 莫三比克 納米比亞 奈及利亞 甘比亞 塞內加爾 南非 坦尚尼亞 多哥 烏干達 尚比亞 辛巴威 盧安達 象牙海岸 加彭

歐洲地區（計30國）

亞美尼亞 亞塞拜然 白俄羅斯 捷克 愛沙尼亞 喬治亞 匈牙利 義大利 希臘 立陶宛 摩爾多瓦 波蘭 波士尼亞與赫塞哥維納 俄羅斯 烏克蘭 保加利亞 比利時 斯洛伐克 塞爾維亞 拉脫維亞 德國 北馬其頓共和國 羅馬尼亞 克羅埃西亞 瑞典 蒙特內哥羅 科索沃 阿爾巴尼亞 西班牙 芬蘭

亞洲地區（計20國）

中國大陸 蒙古 越南 柬埔寨 北韓 寮國 緬甸 菲律賓 韓國 東帝汶 印尼 印度 馬來西亞 不丹 泰國 尼泊爾 新加坡 孟加拉 斯里蘭卡 臺灣

大洋洲地區（計1國）

巴布亞紐幾內亞

美洲地區（計2國）

多明尼加共和國
海地

https://asf.aphia.gov.tw/ws.php?id=17888

What happened to ASF outbreaks in Asia?

- **ASF spread to 20 countries**
  - Very quick spread in 2018-2019
- **ASF affected mainly small holders/producers**
  - Huge number, low biosecurity
- **ASF is now endemic in most countries**
  - Virus keeps circulating among small producers
- **ASF came into wild boar populations**
  - Malaysia, Philippines, Laos, and Vietnam
- **ASF vaccines are now used in several Asian countries, including Vietnam**
  - Live vaccine attenuated by gene deletion



African swine fever (ASF) situation update in Asia & Pacific

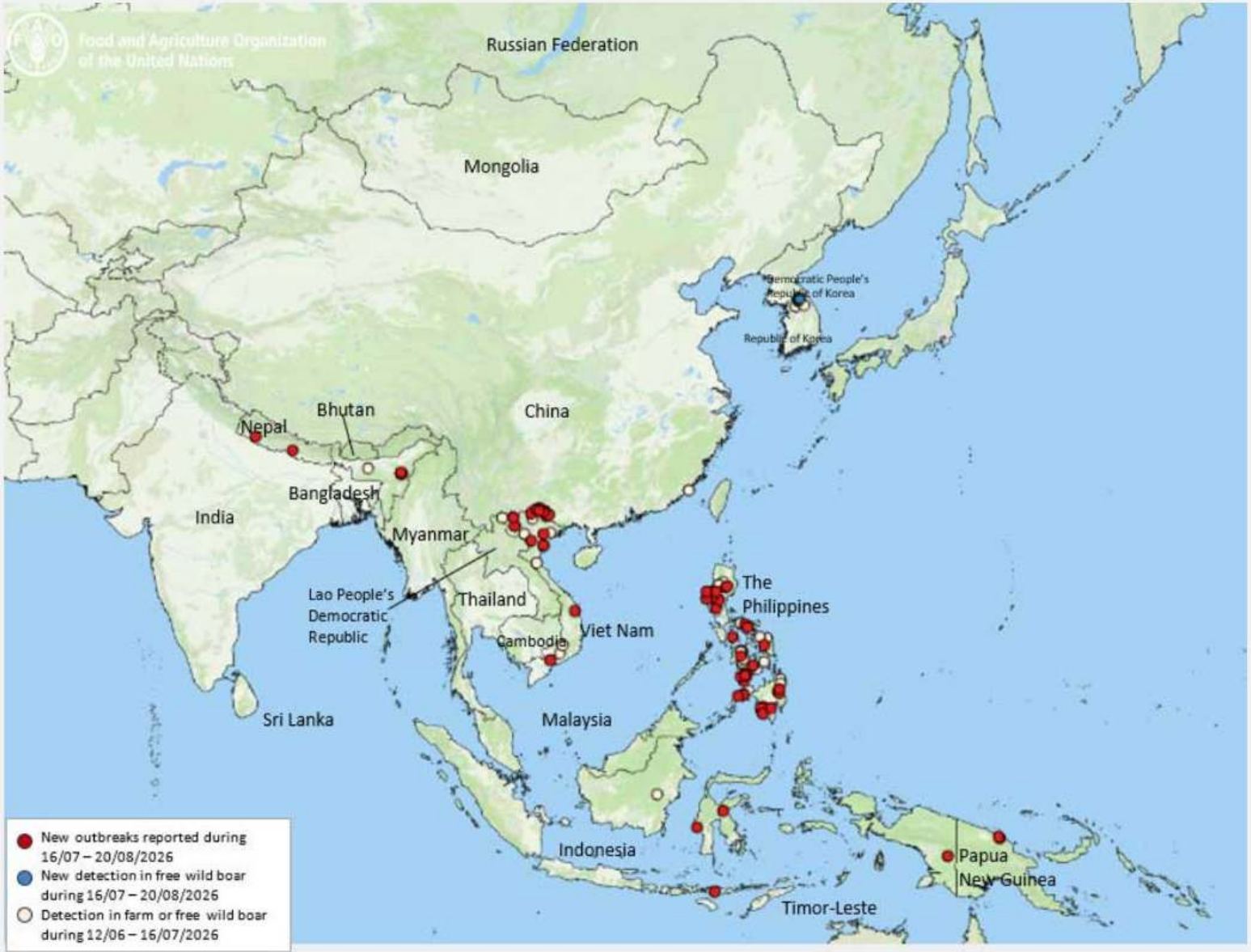
20 August 2026, 08:30 hours, Rome

Overview

Hazard: African swine fever (ASF) is a viral disease affecting pigs and wild boar with up to 100% case fatality rate.

ASF reported countries since August 2018: China, Mongolia, Viet Nam, Cambodia, Democratic People's Republic of Korea, Lao People's Democratic Republic, Myanmar, The Philippines, Republic of Korea, Timor-Leste, Indonesia, Papua New Guinea, India, Malaysia, Bhutan, Thailand, Nepal, Singapore, Bangladesh and Sri Lanka.

Map 1. *ASF situation in Asia (for the past 10 weeks)*



The impact of ASF

- The impact of ASF can have serious animal health, economic, and international trade consequences.
- Animal Health
  - High morbidity and mortality
  - Highly contagious
- Import and export bans
- Movement restrictions
- Quarantine and depopulation
  - Required for elimination
  - 1971 Cuba: 400,000 pigs
  - 2018 Romania: over 120,000 pigs
- It can become a prolonged epidemic

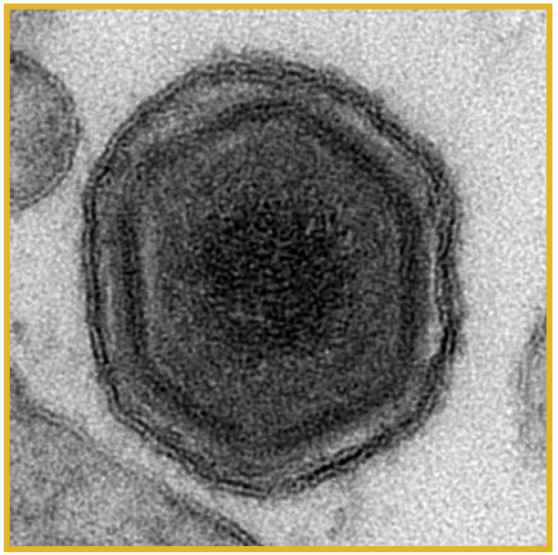
Species affected

- Domestic pigs
- Wild pigs
  - Eurasian wild boars
  - Warthogs (reservoir)
  - Bush pigs (reservoir)
  - Giant forest hogs

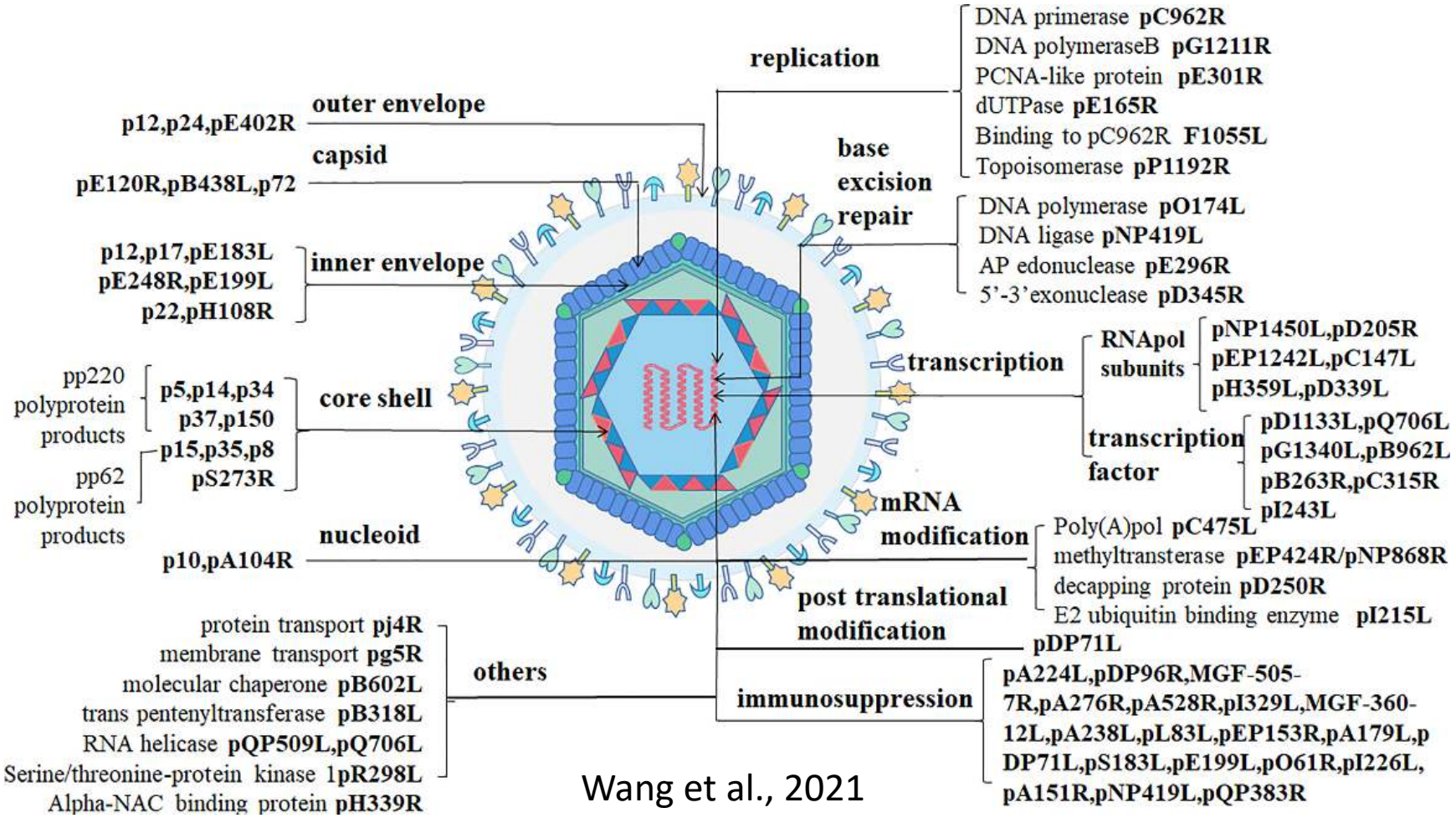


African swine fever virus

- Asfarviridae: *Asfivirus*
  - Large, enveloped DNA virus of 170-193 kb
  - Only arthropod-borne DNA virus



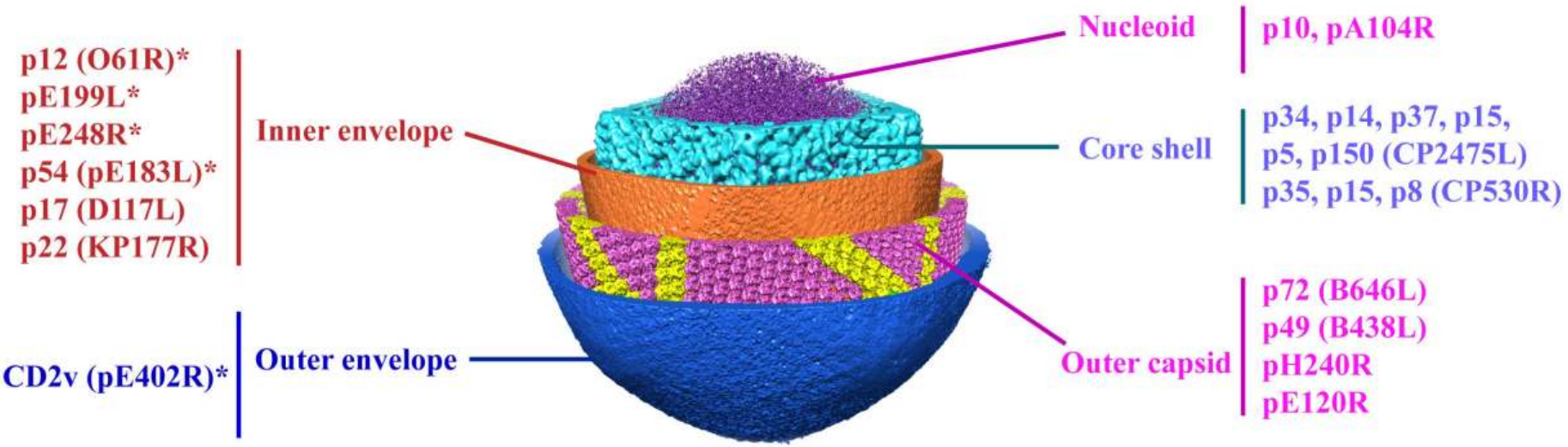
Genome of ASFV



Wang et al., 2021

- Encode more than 160 genes
- Many genes (~1/3) are not essential for virus replication in cells but are important for evading host defenses.

Location of ASFV structural proteins



Zhang et al., 2021

- Complex virus particles containing > 68 proteins in several layers.

Genotypes of ASFV

- 24 genotypes
  - Vary in virulence
  - High virulence: up to 100% mortality
  - Low virulence: seroconversion
  - All genotypes are present in Africa, while genotypes I and II have been found outside of Africa

Resistance of ASFV

- Highly resistant in the environment, especially at lower temperatures
- Survival times
  - Several days in feces at RT
  - Months in contaminated pig pens
  - Up to 18 months in blood at 4°C
  - Over 140 days in some pork products
    - Salted dried hams
  - Years in frozen carcasses

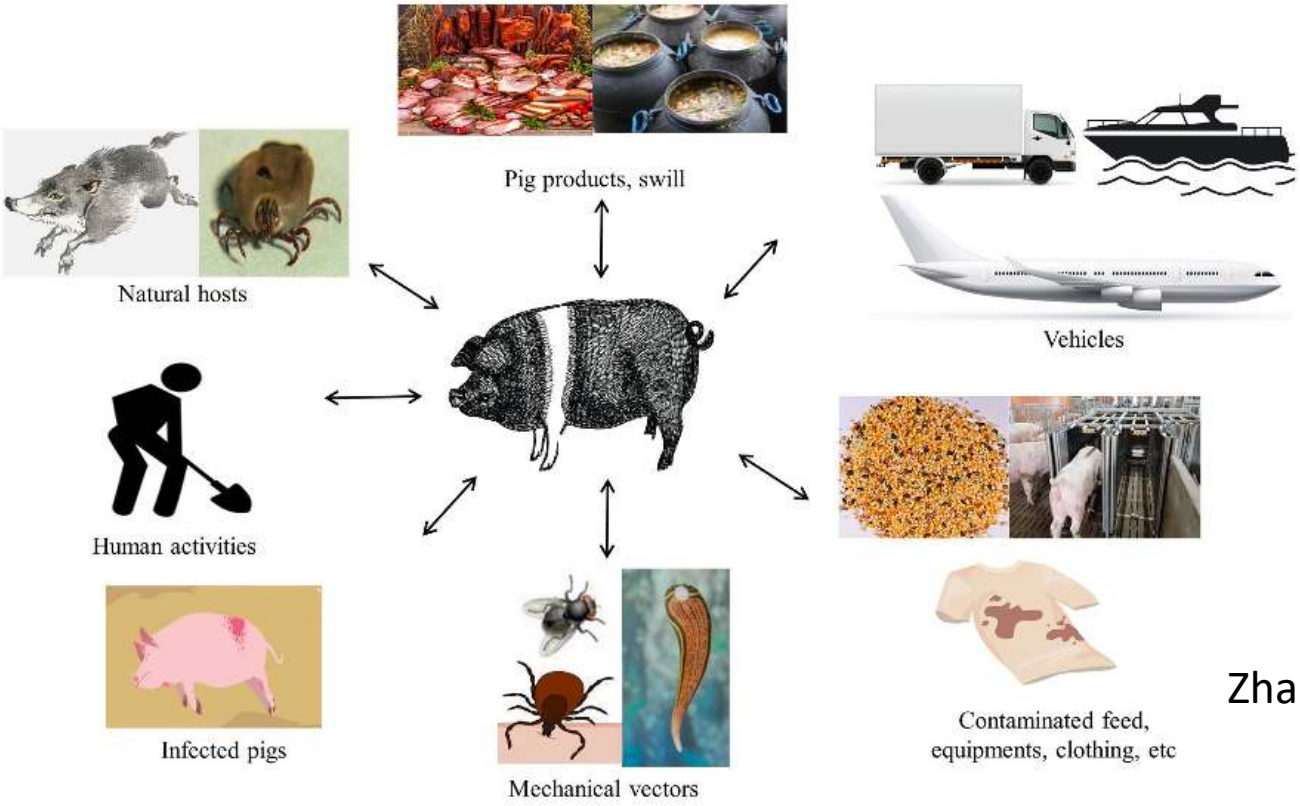
Virus inactivation

- Serum and feces greatly inhibit the efficacy of disinfectants. Thorough cleaning before disinfectant application is important.
- Disinfectants on nonporous surfaces
  - Sodium hypochlorite (次氯酸鈉), citric acid (檸檬酸), some iodine and quaternary ammonium solutions
- Meat/tissue products
  - High temp (70°C/150°F) for 30 min
- Can be inactivated
  - pH below 3.9 or above 11.5
  - A higher pH (13.4) is needed if serum present



https://www.youtube.com/watch?v=eyQ4t1wHl2M

Transmission of ASF



Zhang et al., 2023

- Infected **domestic pigs** and **wild boars**, and contaminated feed, personnel, vehicles, and **pork** are the main sources of ASF infection.

Transmission

- Direct contact with infected pigs
  - Mainly via the oronasal route
  - Virus present in secretions, excretions, tissues, and especially blood
- Ingestion of contaminated pork products
  - Fed to pigs – swill, food waste, and garbage
- Contact with infected carcasses

Transmission

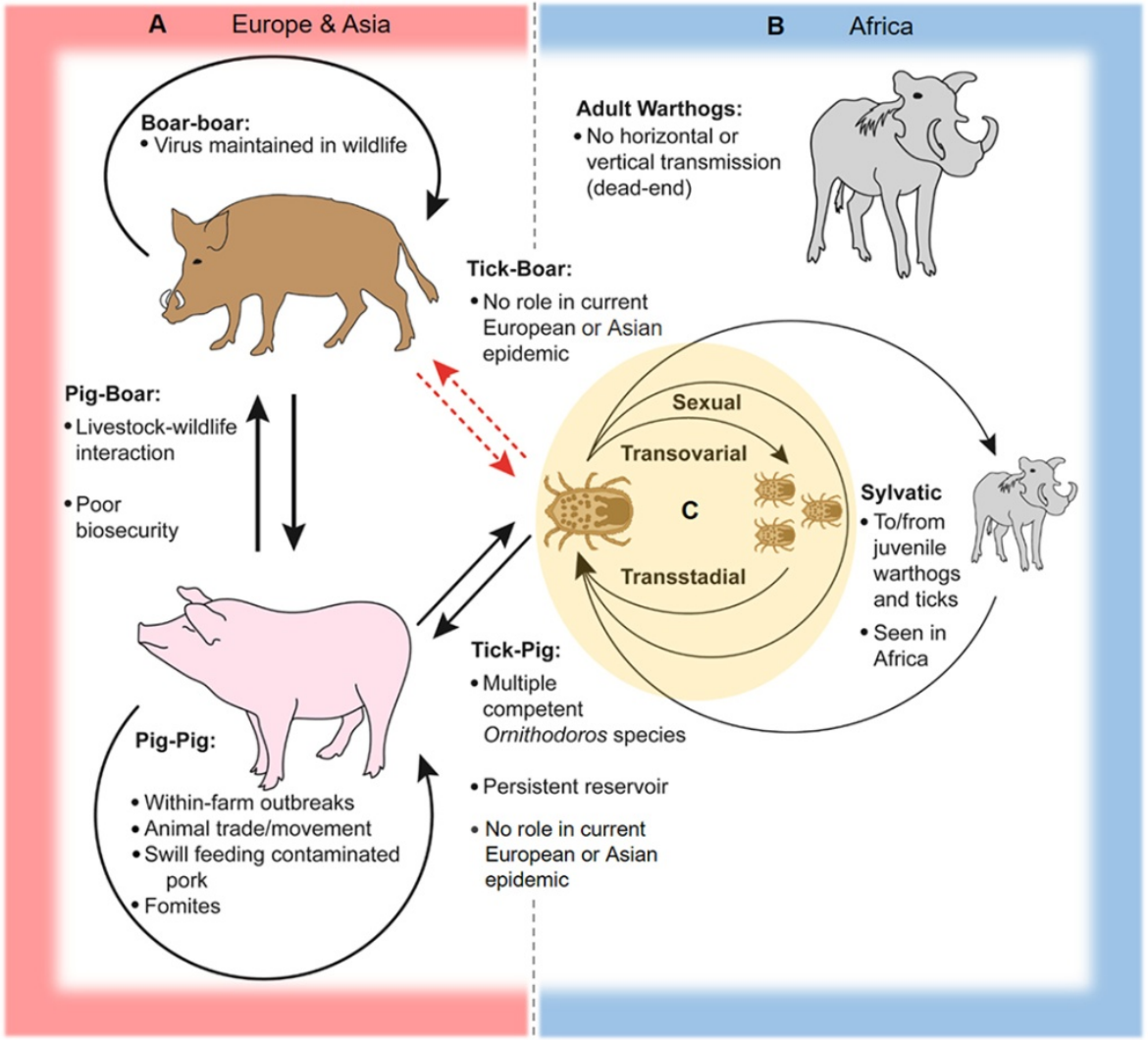
- Fomites
  - Clothing, vehicles, equipment
  - Environmental contamination
    - Blood, diarrhea, feces
- Vectors
  - Biological: Bite from soft ticks *Ornithodoros*
  - Mechanical: Other insects
    - Biting flies (*Stomoxys)*

Transmission

- *Ornithodoros* soft **ticks**
  - Transstadial
  - Transovarial
  - Sexual transmission
  - Infected for life
  - Colonies maintain virus for years
- Tick-to-pig transmission
  - Important in Africa
  - Maintained between warthogs and tick



Transmission cycle



Gaudreault et al., 2020

World Organisation for Animal Health
Founded as OIE

ASF in wild boars



©MARIUS MASIULIS

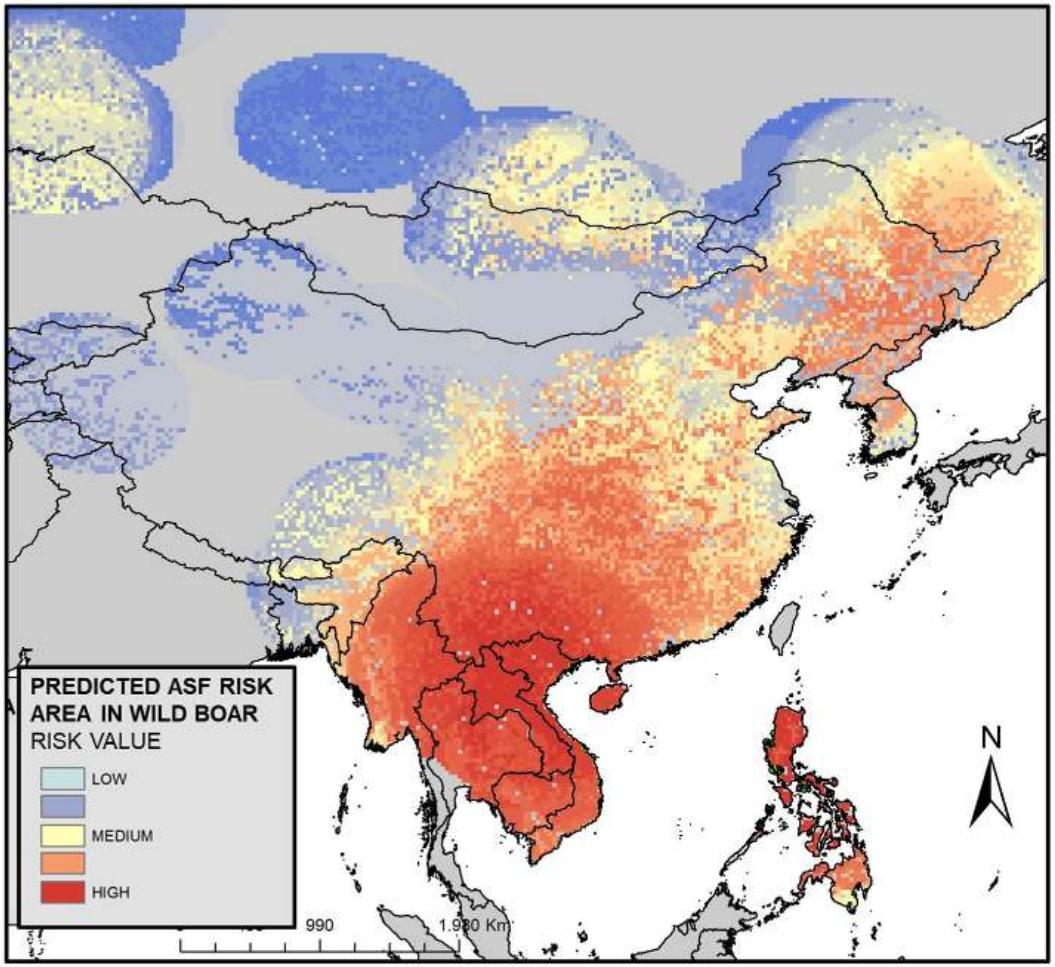
Photo 22
In ASF-infected areas and areas at risk, hunted wild boars should be safely transported to avoid further spread of the virus



©MARIUS MASIULIS

Photo 23
Blood drops contain a very large amount of the virus

ASF in wild boars



Cadenas-Fernández et al., 2022

Pathogenesis

- The sites of primary ASFV replication are the **monocytes** and **macrophages** of the lymph nodes nearest the point of virus entry.
- **Primary replication**: In the case of oral exposure, the monocytes and macrophages of the tonsils and mandibular lymph nodes are the first involved.
- **Secondary replication**: Subsequently, the virus spreads via the blood and/or lymphatic system to sites of lymph nodes, bone marrow, spleen, lung, liver, and kidney.
- Viremia usually begins 4–8 days post-entry and, because of the absence of fully neutralizing antibodies, persists for weeks or months.

Morbidity and Mortality

- Morbidity up to 100%
  - Previously unexposed herds
- Mortality varies with genotype and virulence
  - Ranges from < 5% to 100%
- All ages affected
- Subacute mortality: 30% to 70%
- African wild suids may remain asymptomatic or show mild infection

Clinical signs

- Incubation period
  - 4-19 days following direct contact
  - < 5 days after tick bite

- Forms of disease
  - Peracute – sudden death
    - Maybe the first indication of infection in a herd
  - Acute
  - Subacute
  - Chronic

Clinical signs – Domestic pigs



Fig. 2. Clinical signs of domestic pigs upon infection with highly virulent ASFV strains. First signs are observed app. four days post infection. They include high fever, reluctance to move, inappetence, and huddling (upper row left and center). Some animals develop conjunctivitis and gastro-intestinal signs (vomiting, diarrhea). With progression of the disease, animals become somnolent (lower row left), appear desorientated, and show dyspnea. In the final phase, affected animals may show seizures (upper row right) and haemorrhages (skin haemorrhages in the lower row center, epistaxis lower row right).

Clinical signs – Peracute form

- Loss of appetite
- Body temperature > 41 °C
- Depression
- Cutaneous hyperemia
- Death usually follows 1–4 days after clinical signs appear
- More commonly observed in ASF-naïve populations



Clinical signs – Acute form

- High fever (40–42°C)
- Anorexia
- Lethargy
- Weakness, Recumbency
- Erythema, Cyanosis
- Hemorrhages
  - Skin of ears and flanks



Diseases of Swine, 11th Edition

Clinical signs – Acute form

- Diarrhea
- Abortion
- Respiratory
  - Dyspnea
  - Nasal discharge
- Death
  - Mortality rates range from 90 to 100%
  - 7-10 days after the appearance of clinical signs





Clinical signs – Acute form





Clinical signs – Subacute form

- Moderately virulent isolates
- Similar signs to the acute form, but less severe
  - Abortion
  - Fever
  - Erythema, Cyanosis
- Mortality rates range from 30 to 70%

Clinical signs – Chronic form

- Intermittent, low fever
- Anorexia, depression
- Emaciation, stunting
- Respiratory: coughing
- Joint swelling
- Diarrhea
- Occasional vomiting
- Skin lesions
- May be fatal



Clinical signs – Wild boars

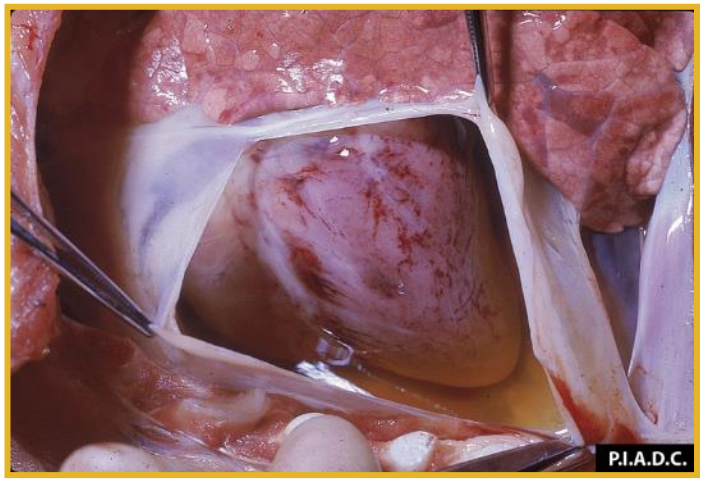
S. Blome, et al. Virus Research 287 (2020) 198099

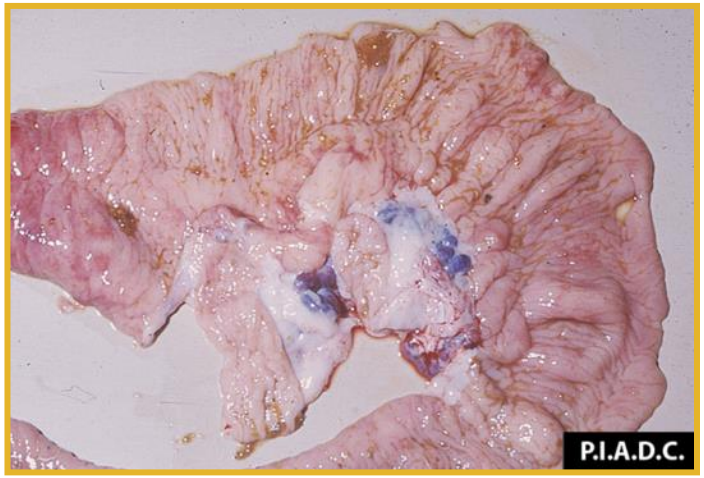


Fig. 3. Clinical signs in wild boar. The signs resemble the courses in domestic pigs. Depression and reduced liveliness are seen in most animal (see upper row left and center, lower row right). The same is true for dyspnea (the animal in the upper row right showed severe respiratory distress). Hind leg paresis (lower row left) and seizures (lower row center) can be observed in the final stage.

Gross lesions

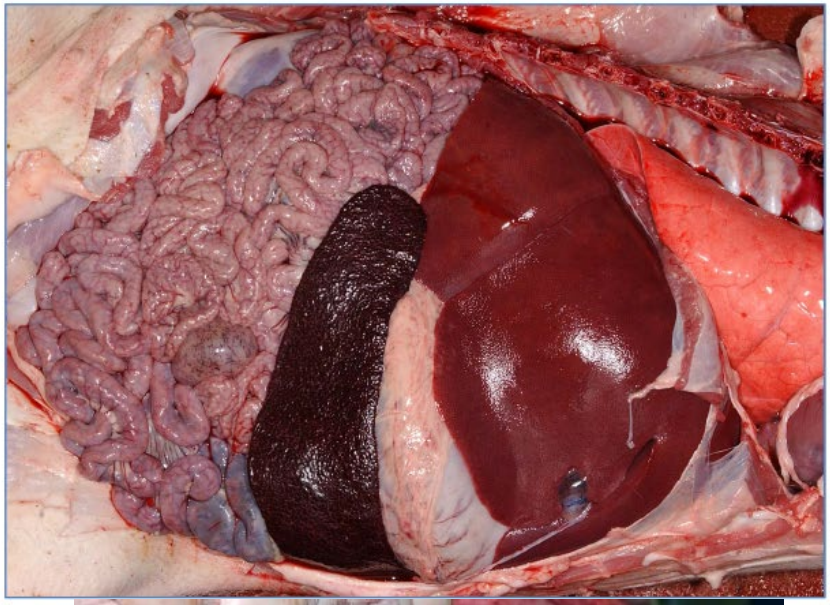
- The principal gross lesions are observed in the spleen, lymph nodes, kidneys, and heart
- Numerous hemorrhagic internal organs
- Skin
  - Discoloration
  - Hemorrhages
- Signs of bloody diarrhea or internal hemorrhages





Gross lesions

- Spleen
  - Enlarged
  - Friable
  - Darkened



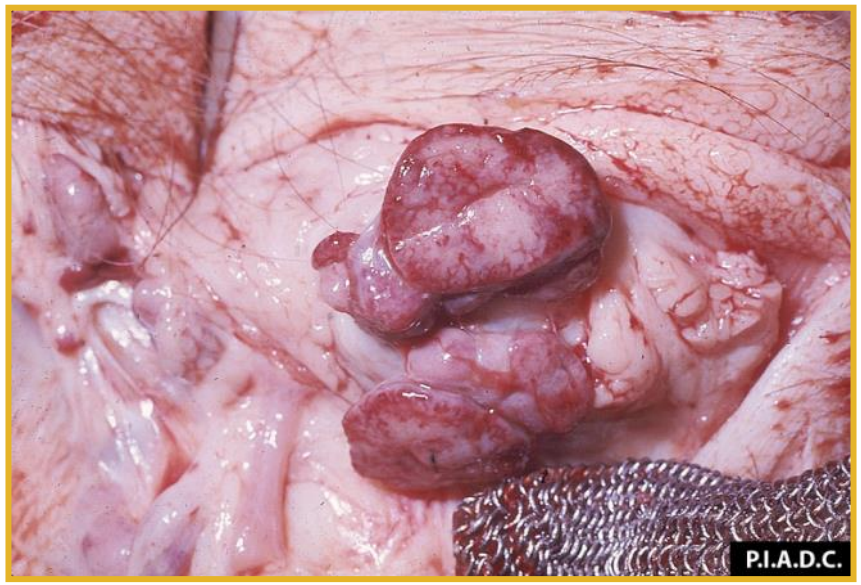


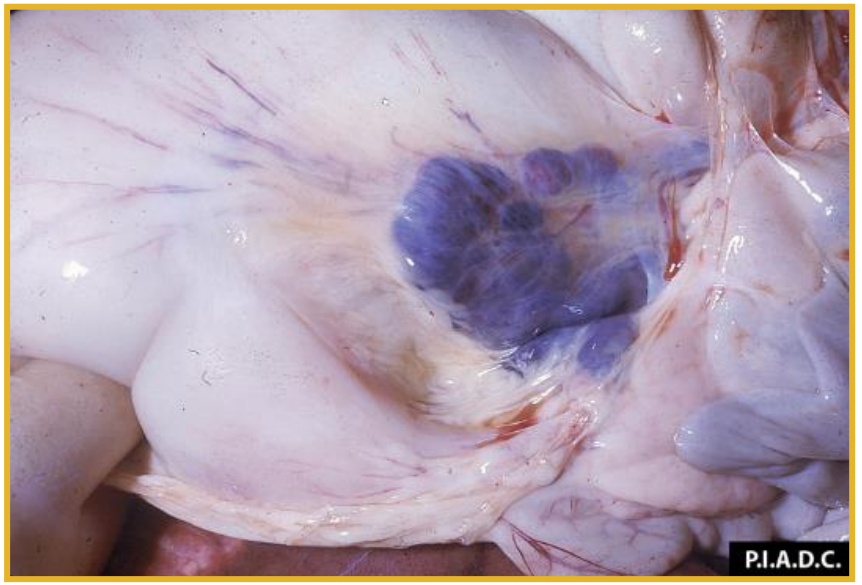




Gross lesions

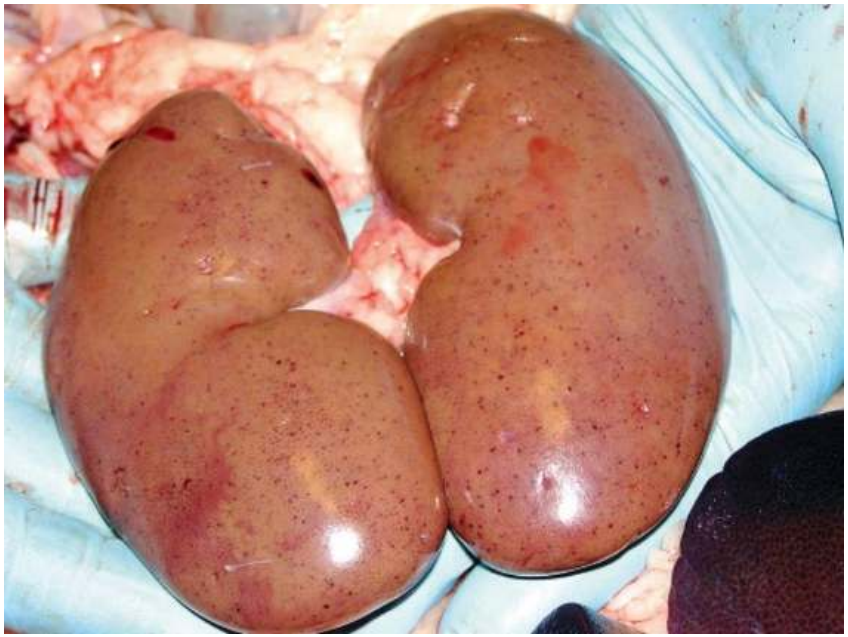
- Lymph nodes
  - Swollen
  - Hemorrhagic





Gross lesions

- Kidneys
  - Petechiae on cortical and cut surfaces
  - Perirenal edema possible

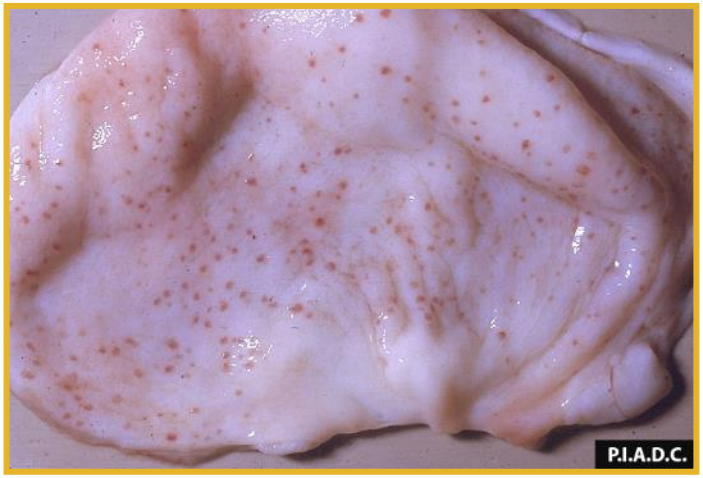


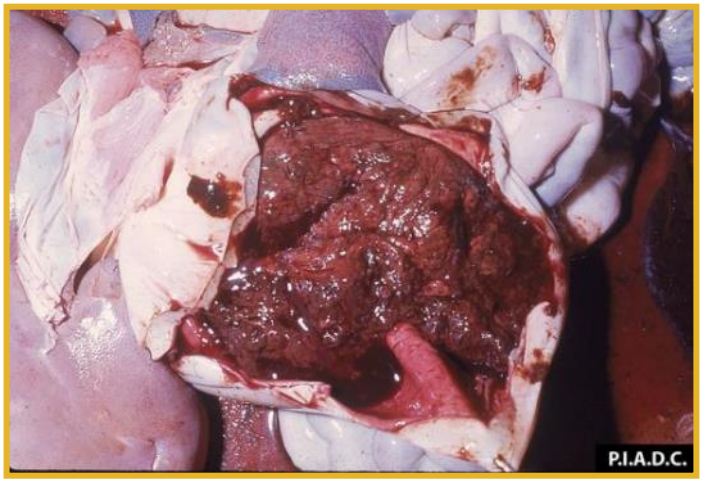


Diseases of Swine, 12th Edition

Gross lesions

- Other organs
  - Hemorrhages
  - Petechiae
  - Ecchymoses
- Edema
  - Lungs
  - Gall bladder
- Brain/meninges
  - Congested
  - Edema



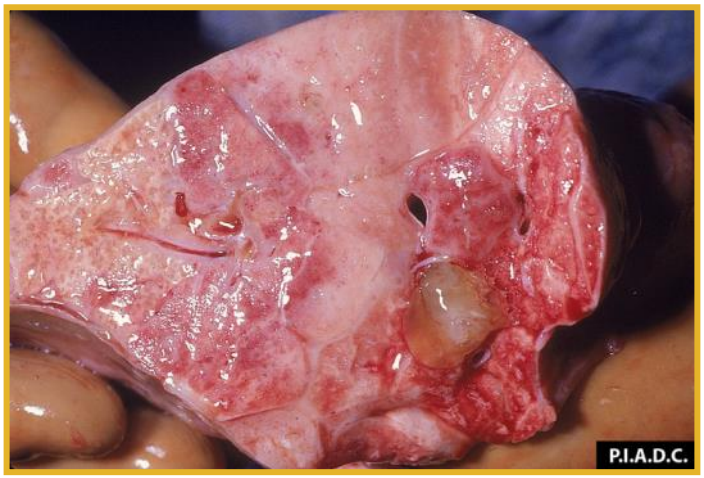


Gross lesions – Chronic Infection

- Focal skin necrosis
- Skin ulcers
- Consolidated lung
- Caseous pneumonia
- Fibrinous pericarditis
- Pleural adhesions
- Lymphadenopathy
- Swollen joints

Focal skin necrosis





Consolidated lung

Differential diagnosis –
Generalized septicemic or hemorrhagic conditions

- Classical swine fever
- Acute PRRS
- Porcine circovirus-associated disease (PCVAD)
- Pseudorabies
- Erysipelas
- Salmonellosis
- Actinobacillosis
- Glasser's disease
- Eperythrozoonosis
- Thrombocytopenic purpura
- Warfarin poisoning
- Heavy metal toxicity

Center for Food Security and Public Health, Iowa State University, 2018

Diagnosis

ASF sample matrices and tests

- Whole blood
  - alive/dead pigs
- Lymph nodes
  - submandibular,
  - Inguinal
  - gastrohepatic
- Spleen
- Lung
- Kidney
- Tonsils
- Bone marrow
- Tonsil scrapings/ swabs

Serological analyses

- ELISA
- Lateral flow device (antibody)
- Immunoblotting test
- Indirect fluorescent antibody test
- Immunoperoxidase test

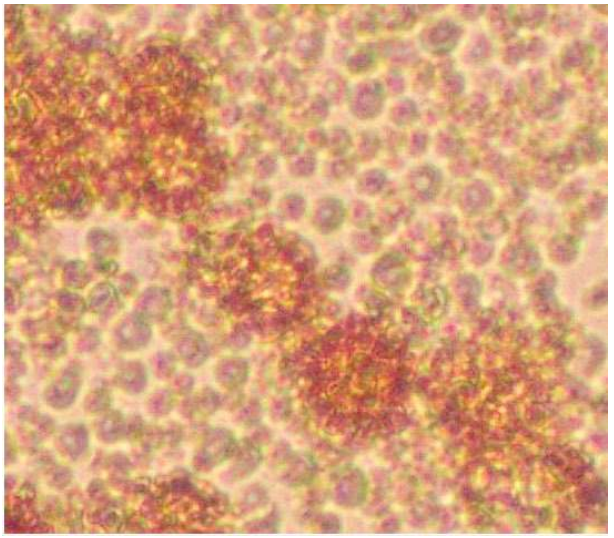
Virological & molecular analyses

- Antigen ELISA
- Lateral flow device (antigen)
- Virus isolation and HAD
- Direct fluorescent antibody test
- PCR and real-time PCR
- Sequencing

Source: Dr. Flannery, Pirbright

Diagnosis – Antigen

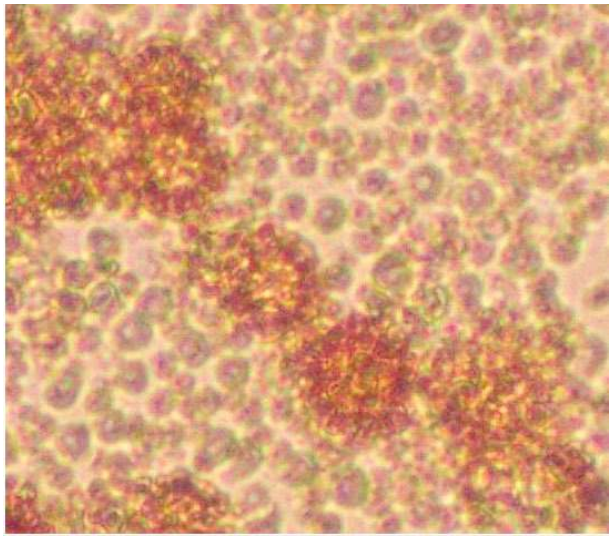
- Virus isolation and HAD
  - ASFV infects and replicates in primary leukocyte cultures from pig peripheral blood
  - Inoculation of sample material on susceptible primary cell cultures of porcine origin: monocytes and macrophage cells
  - ASFV replicates in the cells, and the CPE (haemadsorption and cell lysis) will be produced in the infected cells
  - Rosettes form due to **haemadsorption (HAD)** of porcine erythrocytes on ASFV-infected macrophages



Source: Dr. Flannery, Pirbright

Diagnosis – Antigen

- Virus isolation and HAD
  - The test is highly specific: only ASFV is capable of haemadsorbing in leukocyte cultures
  - Primary cell culture is required
  - Not all ASFV strains haemadsorb
  - Test duration is 3-10 days
  - Virus isolation and identification by HAD are recommended as a reference test for the confirmation of positive results



Source: Dr. Flannery, Pirbright

Diagnosis – Antigen

- PCR
  - Clinical samples, putrefied samples, fresh tissues/blood





Diagnosis – Antibody

- Serological
  - ELISA
  - Screening assays

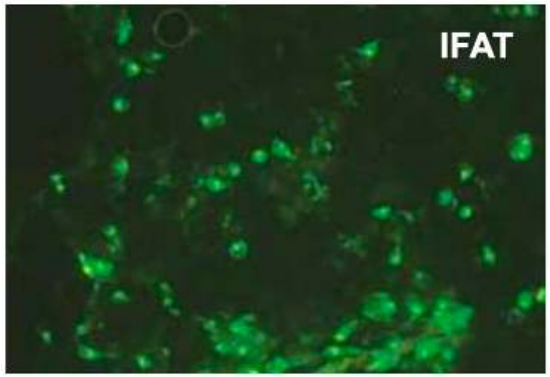


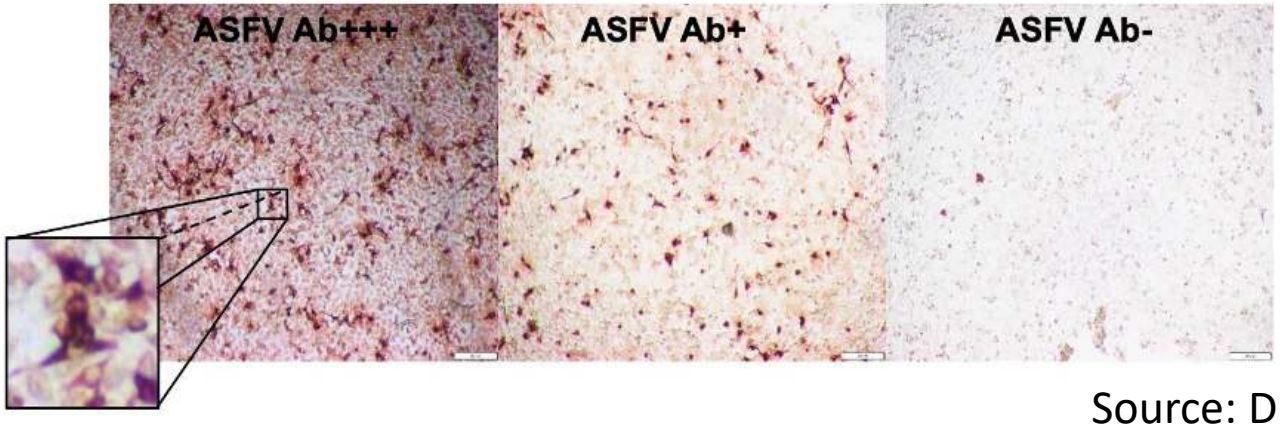


Diagnosis – Antibody

- Serological
  - Immunoblotting test (IBT)
  - Indirect fluorescent antibody test (IFAT)
  - Indirect immunoperoxidase (IPT)
  - Confirmatory assays







Source: Dr. Flannery, Pirbright

Prevention and Control



- ASF vaccines are now licensed and used in some countries, but no vaccine is yet widely accepted as a global solution for ASF control
  - **Gene-deletion live-attenuated** vaccines are currently the most advanced approach, but safety and efficacy require careful evaluation
- Prevention in ASF-free countries relies on implementing appropriate **import policies** and **strict biosecurity measures** on farms, ensuring that neither infected live pigs nor pork products are introduced into ASF-free areas.
- **Surveillance** for early detection

Prevention

Prevent **direct** transmission

- Isolate ill pigs
- Prevent contact with wild boars; when possible, house pigs indoors
- Keep newly acquired pigs separate from the herd for at least 30 days to ensure their health

Prevention

Prevent **indirect** transmission

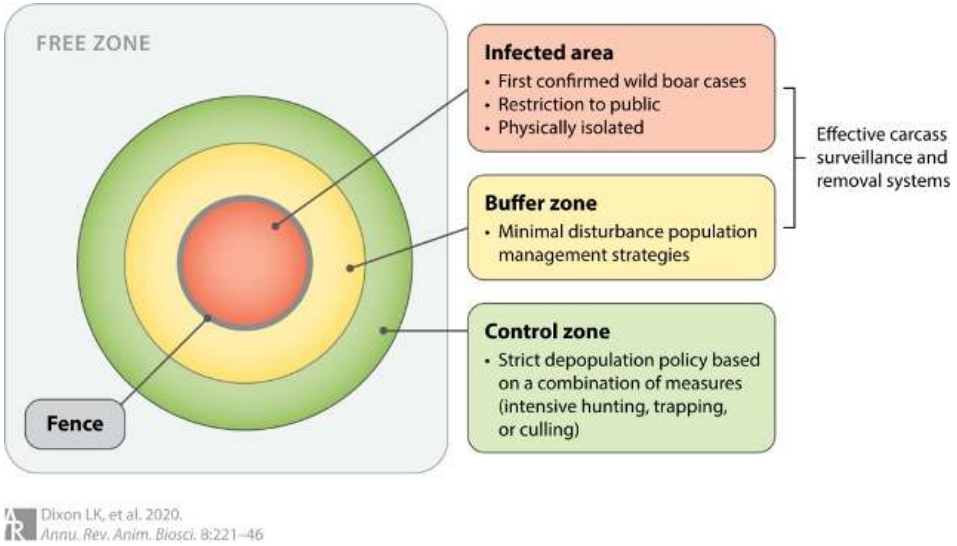
- Do not feed uncooked pork products to pigs
  - Swill, garbage, waste
- Disinfection
  - Vehicles, equipment, footwear, clothing
- Appropriate disposal of manure and carcasses
- Avoid hunting wild boars prior to contact with domestic pigs

Prevent **vector** transmission

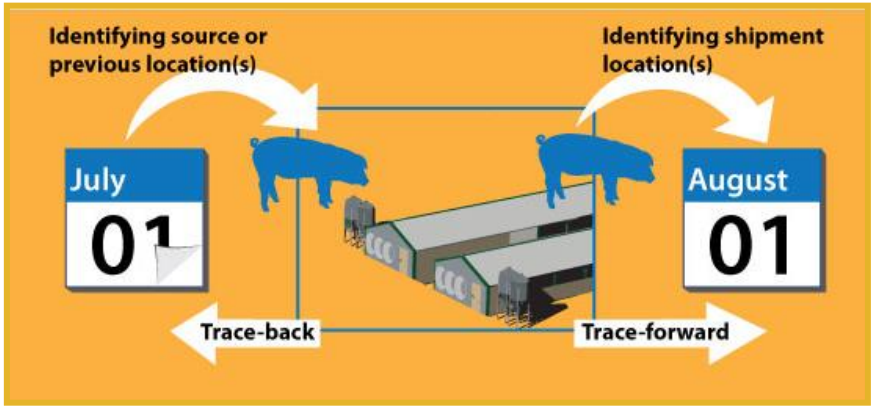
- Control tick and other insect vectors
- May be difficult in endemic areas

Control measures

- Quarantine of the farm
- Authorities notified
- Movement restrictions
- Investigation
- Rapid diagnosis
- Determine source of virus





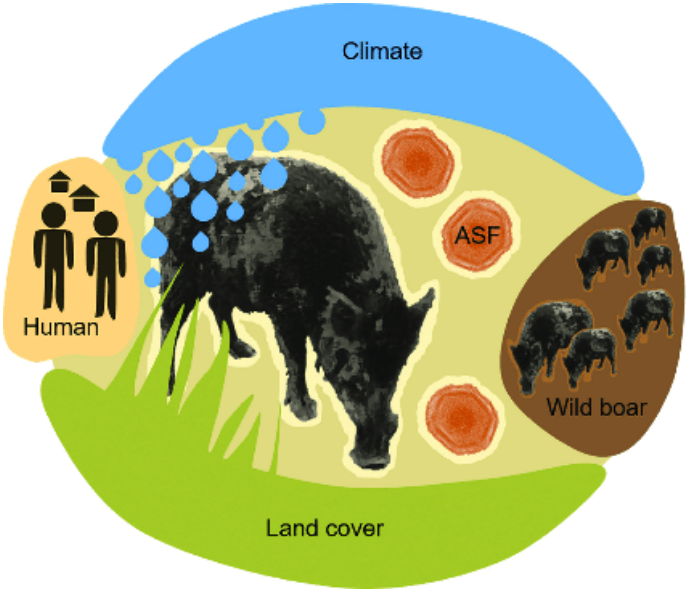


Elimination

- Euthanize/depopulation of infected and in-contact animals
- Proper carcass disposal
  - To avoid further spread
- Sanitation/disinfection
- Movement controls/quarantines
- Avoid contact with wild boars

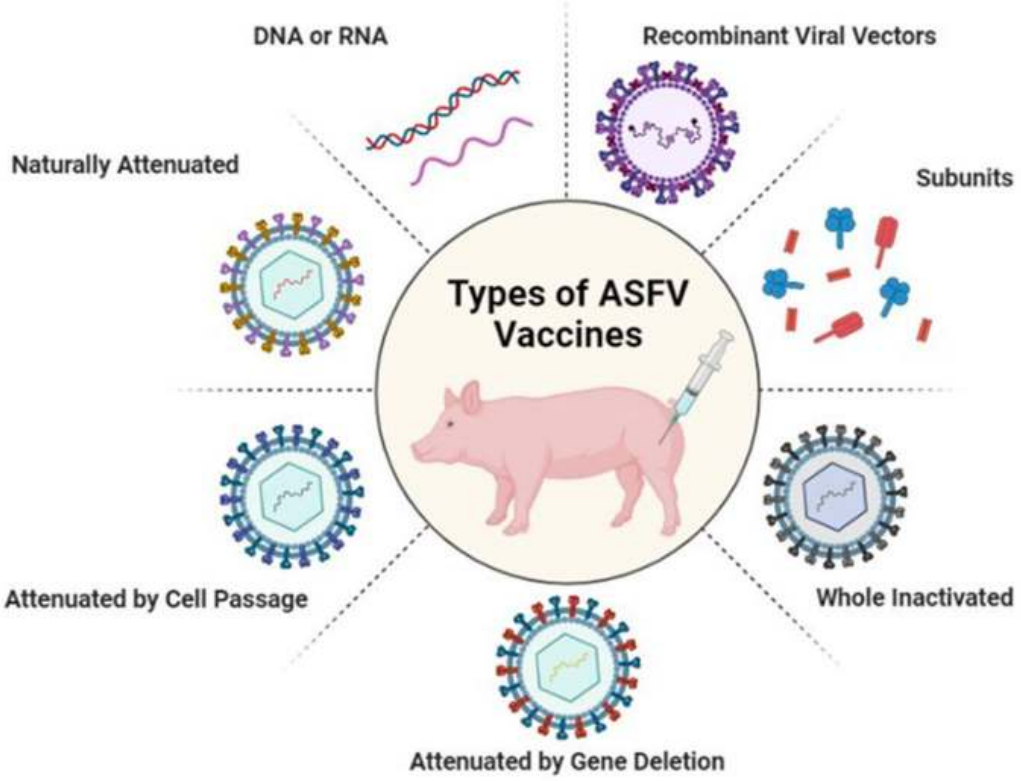
Prevention and Control – Wild boars

- As observed in Europe and some regions of Asia, **wild boars** play a significant role in the transmission of ASF.
- Managing wild boar populations and minimizing interactions with low-biosecurity pig production systems are critical.
- Effective surveillance and management of wild boar populations are essential for ASF prevention and control.



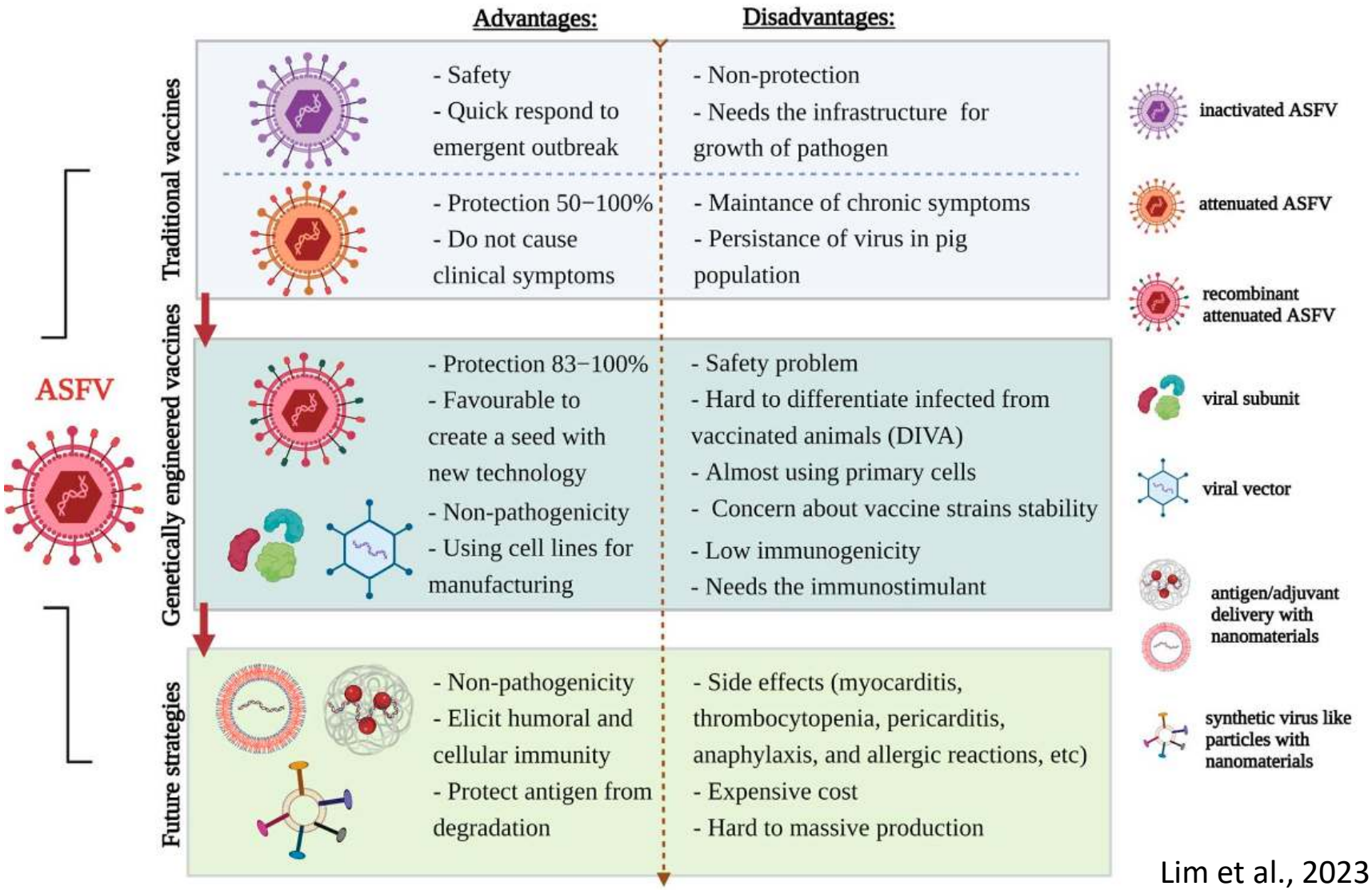
ASFV vaccine candidates

- **Gene-deletion live-attenuated** vaccines are among the most promising ASF vaccine approaches.



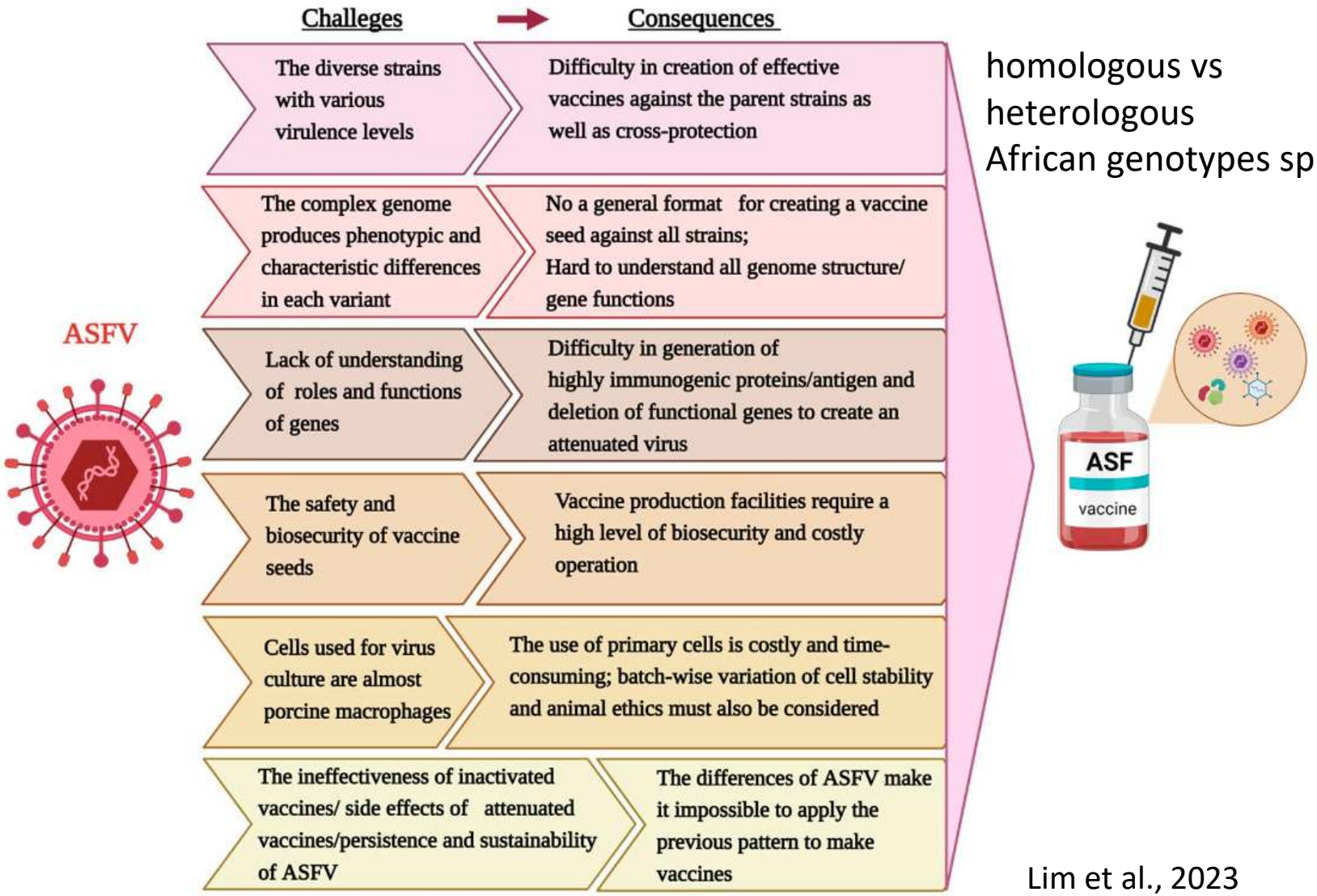
Chathuranga et al., 2023

The strategies for ASF vaccine development



Lim et al., 2023

Challenges in ASF vaccine development



homologous vs
heterologous
African genotypes spreading

Lim et al., 2023

Commercialized vaccines in Vietnam

NAVETCO



Trade name: **NAVET-ASFVAC**

- ASFV-G-ΔI177L Strain
- Cell line: PBMC
- Virus titer ≥ $10^{2,6}HAD_{50}$/dose

1st licensed in 2022

AVAC



Trade name: **AVAC ASF Live**

- ASFV-G-ΔMGF Strain
- Cell line: DMAC
- Virus titer ≥ $10^{3,5}HAD_{50}$/dose

2nd licensed in 2023

DABACO



Trade name: **DACOVAC ASF2**

- ASFV-G-ΔI177L/ΔLVR Strain
- Cell line: PIPEC
- Virus titer ≥ $10^{3,0}HAD_{50}$/dose

3rd licensed in 2025

https://rr-asia.woah.org/app/uploads/2023/09/8-5-vietnam-asf_vaccines.pdf

Current use of ASF vaccines

- Vietnam
  - First country to approve a commercial ASF vaccine in June 2022
  - Three licensed live-attenuated vaccines
  - Vaccination incorporated into national ASF control strategies
- Philippines
  - Controlled use of AVAC ASF LIVE initiated before commercial registration
  - Approved for commercial sale in September 2026
  - Currently approved for use in fattening pigs
- Indonesia
  - Emergency use of AVAC ASF LIVE approved in April 2025
  - Vaccine importation and deployment under government oversight

ASFV surveillance in Africa

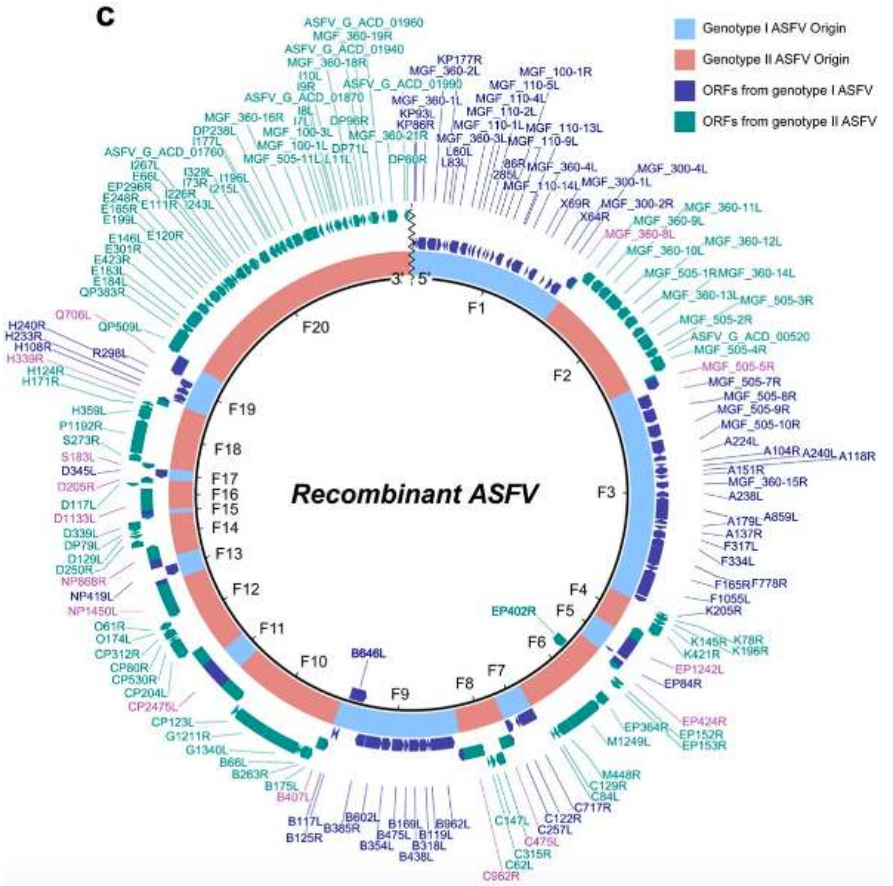
- Conduct surveillance for various genotypes in wild boars and ticks in Africa.
- Identify any emergence of new genotypes or the potential spread of existing genotypes outside Africa.



Photos: Dr. Gladue, USDA

Highly lethal genotype I and II recombinant ASFV detected in pigs

- These recombinants are classified as **genotype I** according to their ***B646L*** gene, yet 10 discrete fragments are derived from **genotype II** virus.
- The live attenuated vaccine derived from **genotype II** ASFV is **not** protective against the challenge of the recombinant virus.
- These naturally occurring recombinants of **genotype I** and **II** ASFVs have the potential to pose a challenge to the global pig industry.

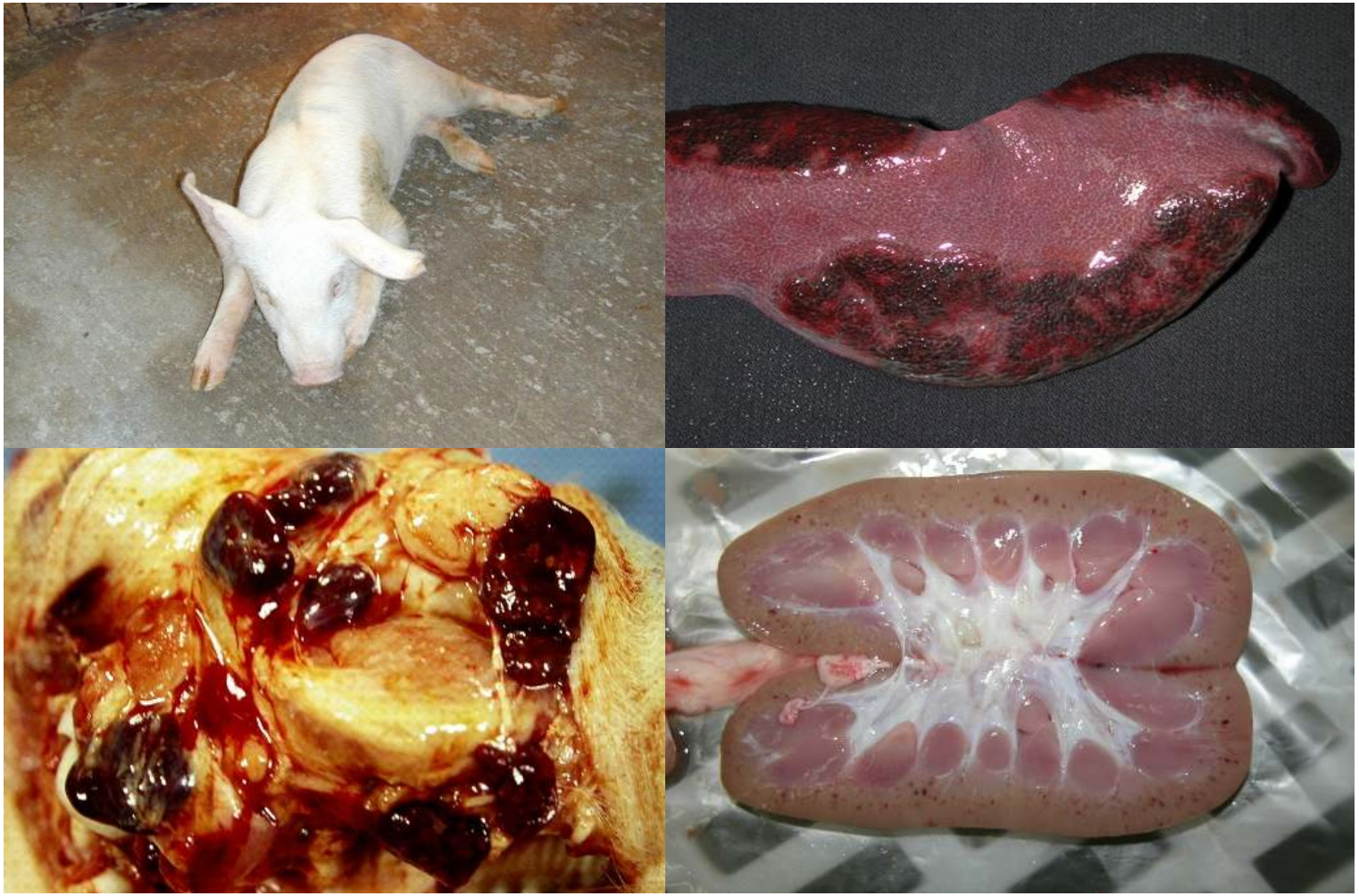


Zhao et al., 2023

Take-home messages

- ASF is a severe, non-zoonotic disease that can cause very high mortality and major economic and trade losses.
- Environmental persistence and human-mediated movement of contaminated pork, fomites, and vehicles are major challenges for ASF control.
- Wild boars can maintain ASFV circulation and make eradication much more difficult.
- Early detection, rapid diagnosis, movement control, depopulation, and strict biosecurity remain the basis of ASF control; vaccination is an emerging additional tool.

Classical swine fever



Classical swine fever

- Classical swine fever (CSF) is a highly contagious disease in pigs, often resulting in the death of infected pigs.
- Fever and bleeding in infected pigs are the main characteristics of classical swine fever.
- In Taiwan, it is classified as a Class A contagious disease.

CSF infected animals



Domestic pigs



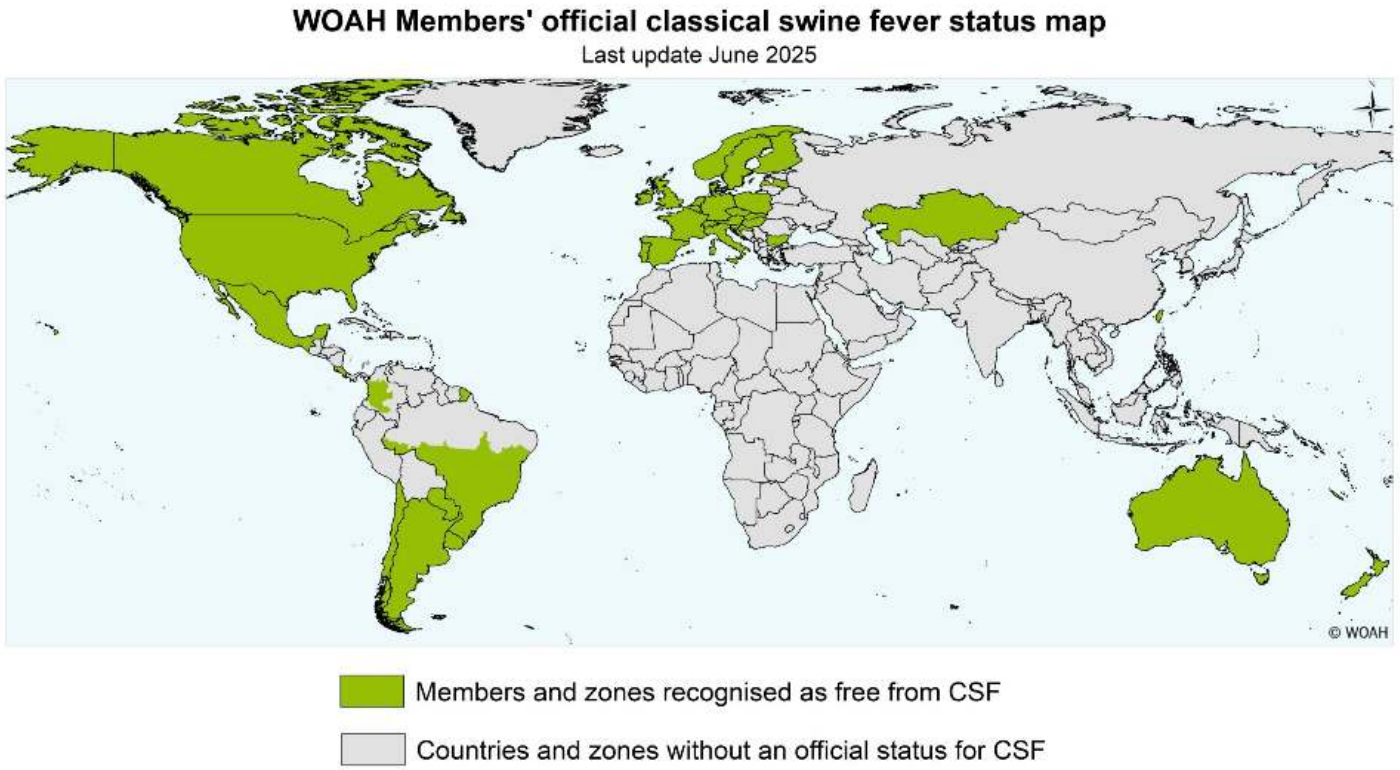
Wild boars

Classical swine fever

- CSF infection can have a mortality rate as high as 100%.
- The younger the pigs are infected, the higher the mortality rate.
- Symptoms of CSF include high fever, depression, neurological symptoms, and abortion.
- Infection with CSF can cause immunosuppression in pigs.

Global situation of CSF – Free countries

WOAH, June, 2025



- A total of 39 member countries are recognized as free from CSF: Taiwan, the United States, Canada, New Zealand, Australia, and most of Europe.

The current global situation of CSF – Free zones

CSF free zones in Brazil



Official CSF status in Brazil

- CSF free zone, consisting of the States of Santa Catarina and Rio Grande do Sul (September 2014)
- CSF free zone, consisting of the States of Acre, Bahia, Espirito Santo, Goias, Mato Grosso, Mato Grosso do Sul, Minas Gerais, Rio de Janeiro, Rondônia, São Paulo, Sergipe and Tocantins, Distrito Federal, and the municipalities of Guajará, Boca do Acre, South of the municipality of Canutama and Southwest of the municipality of Lábrea in the State of Amazonas (September 2015 and October 2020)
- CSF free zone, consisting of the State of Paraná (October 2020)
- Zone of Brazil without a recognised CSF status
- State: Framed State is partly included in the CSF free zone

* Dates shown in brackets indicate when the relevant applications were submitted to the OIE by the Delegate

CSF free zones in Colombia

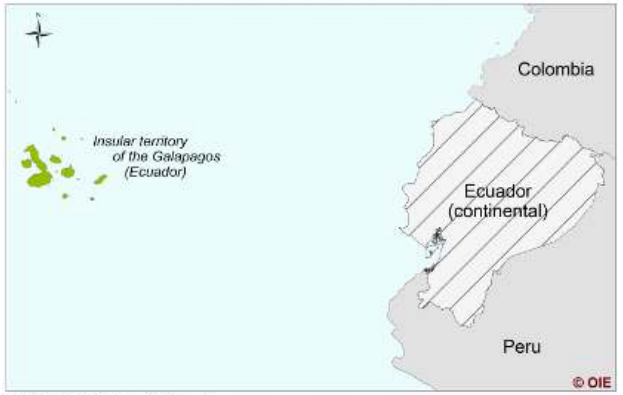


Official CSF status in Colombia

- CSF free zone, consisting of the departments of Antioquia (with the exception of the Magdalena Medio, Uraba and Lower Cauca), Caldas (with the exception of the Magdalena Medio), Quindio, Risaralda, Valle del Cauca, northern zone of the Cauca, Choco and the municipality of Cajamarca in Tolima (September 2015)
- CSF free zone, consisting of the municipalities of Nariño, Puerto Berrio, Puerto Nare, Puerto Triunfo, Caucasia, Valdivia and Yondó in the department of Antioquia; the municipalities of La Dorada, Manzanares, Marquetalia, Norcasia, Pensilvania, Samaná and La Victoria in the department of Caldas; the municipalities of Chámeza, La Salina, Monterrey, Recetor, Sácama, Sabanalarga, Tauramena and Villanueva in the department of Casanare; the departments of Boyacá (with exception of the municipality of Cubará), Caquetá, Cauca (with exception of the municipalities in the other free zone and Argelia, Balboa, Florencia and Mercaderes), Cundinamarca, Huila, Meta, Santander and Tolima (with exception of the municipality of Cajamarca) (October 2020)
- Zone of Colombia without a recognised CSF status
- Departments: Framed departments are partly included in the CSF free zones

* Date shown in brackets indicates when the relevant application was submitted to the OIE by the Delegate

CSF free zone in Ecuador



Official CSF status in Ecuador

- CSF free zone, consisting of the Insular territory of the Galapagos (October 2018)
- Zone of Ecuador without a recognised CSF status

* Dates shown in brackets indicate when the relevant application was submitted to the OIE by the Delegate

WOAH, 2025

The impact of CSF

- Between 1997 and 1998, CSF broke out in the Netherlands.
- Over 400 pig farms were infected in total.
- Approximately 12 million pigs were culled.
- The losses exceeded 2.3 billion US dollars.

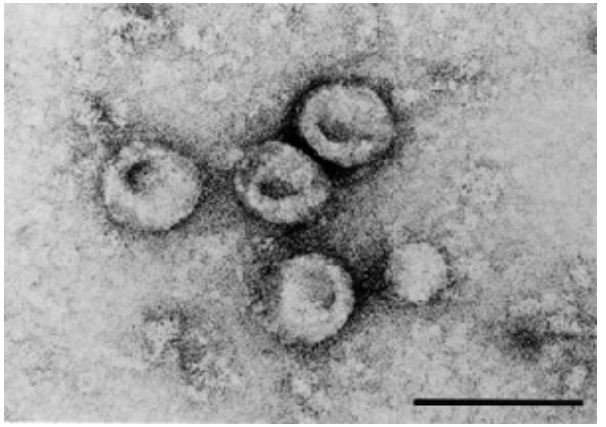


Incubation period

- Incubation period: 2 to 15 days
- Clinical phase
  - Acute: 10 to 15 days
  - Chronic/Congenital/Postnatal: Several weeks to months

Classical swine fever virus

- CSFV is an enveloped RNA virus.
- It belongs to the genus *Orthopestivirus* of the family *Pestiviridae*.
- CSFV has a single serotype.
- There are three genotypes of CSFV.



Taxonomy of Pestiviruses

Species	Virus Name	Abbreviation [1]	Host Species
Pestivirus A	bovine viral diarrhea virus-1	BVDV-1	*Bos* sp., *Ovis* spp., *Capra* spp., *Artiodactyla*
Pestivirus B	bovine viral diarrhea virus-2	BVDV-2	*Bos* sp., *Ovis* spp., *Capra* spp., Artiodactyla
Pestivirus C	classical swine fever virus	CSFV	*Sus scrofa*
Pestivirus D	border disease virus	BDV	*Ovis* spp., *Capra* spp., *Artiodactyla*
Pestivirus E	pronghorn antelope pestivirus	PHV	*Antilocapra americana*
Pestivirus F	porcine pestivirus	BuPV	*Sus scrofa*
Pestivirus G	giraffe pestivirus	-	*Bos taurus, Giraffa camelo-pardalis*
Pestivirus H	HoBi-like pestivirus	HoBi	*Bos* sp.
Pestivirus I	Aydin-like pestivirus	-	*Ovis* spp., *Capra* spp., *Bos taurus*
Pestivirus J	rat pestivirus	NrPV	*Rattus norvegicus*
Pestivirus K	atypical porcine pestivirus	APPV	*Sus scrofa*

Proposed Species	Virus Name	Abbreviation [1]	Host Species
Pestivirus L	Linda virus	LindaV	*Sus scrofa*
Pestivirus M	Phocoena pestivirus	PhoPeV	*Phocoena phocoena*
Pestivirus N	Tunisian sheep-like pestivirus	TSV	*Capra aegagrus hircus Ovis gmelini aries*
Pestivirus O	ovine/IT pestivirus	ovIT PeV	*Ovis gmelini aries*
Pestivirus P	pangolin pestivirus	DYPV	*Amblyomma javanense Manis javanica*
Pestivirus Q	rodent pestivirus	RtNn-PeV	*Niviventer niviventer*
Pestivirus R	rodent pestivirus	RtAp-PeV	*Apodemus peninsulae*
n.a.	rodent pestivirus	RtNe-PeV	*Niviventer excelsior*
n.a.	rodent pestivirus	RtAd-PeV	*Apodemus draco*
n.a.	rodent pestivirus	RtNn-PeV	*Niviventer niviventer*
Pestivirus S	bat pestivirus	BtSk-PeV	*Scotophilus kuhlii*
n.a.	bat pestivirus		
n.a.	bat pestivirus		

Postel *et al*., 2021

Taxonomy of Pestiviruses

Table 4 Changes to species names in the genus *Pestivirus*

Previous species name (if applicable)	New species name[¶]	Virus name (abbreviation)[&]
Pestivirus A	*Pestivirus bovis*	bovine viral diarrhea virus 1 (BVDV1)
Pestivirus B	*Pestivirus tauri*	bovine viral diarrhea virus 2 (BVDV2)
Pestivirus C	*Pestivirus suis*	classical swine fever virus (CSFV)
Pestivirus D	*Pestivirus ovis*	border disease virus (BDV)
Pestivirus E	*Pestivirus antilocaprae*	pronghorn antelope pestivirus (PAPeV)
Pestivirus F	*Pestivirus australiaense*	Bungowannah virus (BuPV)
Pestivirus G	*Pestivirus giraffae*	giraffe pestivirus (GPeV)
Pestivirus H	*Pestivirus brazilense*	HoBi-like pestivirus (HoBiPeV)
Pestivirus I	*Pestivirus aydinense*	Aydin-like pestivirus (AydinPeV)
Pestivirus J	*Pestivirus ratti*	rat pestivirus (RPeV)
Pestivirus K	*Pestivirus scrofae*	atypical porcine pestivirus (APPeV)
/	*Pestivirus L* (proposed to be renamed as "*Pestivirus austriaense*")	Linda virus (LindaV)
/	*Pestivirus M* (proposed to be renamed as "*Pestivirus phocoenae*")	Phocoena pestivirus (PhoPeV)
/	*Pestivirus N* (proposed to be renamed as "*Pestivirus tunisiaense*")	Tunisian sheep-like pestivirus (TSV)
/	*Pestivirus O* (proposed to be renamed as "*Pestivirus italiaense*")	ovine/IT pestivirus (ovIT PeV)
/	*Pestivirus P* (proposed to be renamed as "*Pestivirus pholidotae*")	pangolin pestivirus (DYPV)
/	*Pestivirus Q* (proposed to be renamed as "*Pestivirus niviventris*")	rodent pestivirus (RtNn-PeV)
/	*Pestivirus R* (proposed to be renamed as "*Pestivirus apodemi*")	rodent pestivirus (RtAp-PeV)
/	*Pestivirus S* (proposed to be renamed as "*Pestivirus scotophili*")	bat pestivirus (BtSk-PeV)

Genotypes of CSFV

- CSFV is classified into three genotypes
  - Genotype I: includes ancient virus strains and traditional live attenuated vaccine strains.
  - Genotype II: includes virus strains that have been prevalent since the 1980s.
  - Genotype III: includes virus strains that are unique to specific geographic regions, such as Taiwan, South Korea, Japan, Thailand, and the United Kingdom.
  - In European and Asian countries, CSFV strains circulating in the field have shifted from the original genotype I or genotype III to genotype II.

Virulence of CSFV

- High virulent
  - The course of the disease is acute, and nearly all infected pigs will die.
- Moderate virulent
  - Subacute or chronic infection
- Low virulent
  - Subclinical infection, typically pathogenic to fetuses
- Avirulent: Vaccine strain
  - Avirulent

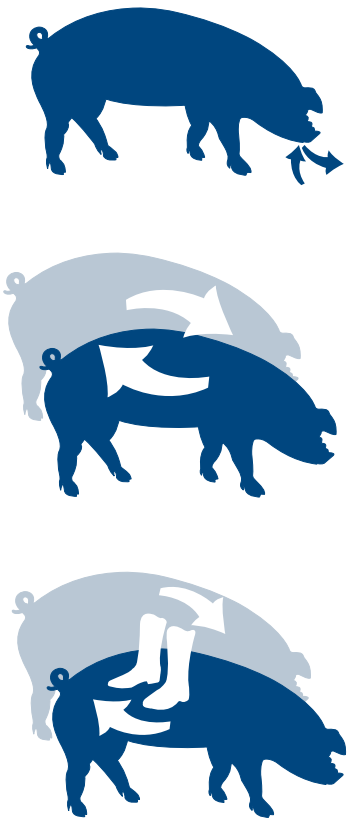
Resistance of CSFV

- Temperature sensitivity
  - Survives for six weeks at 4°C
  - Survives for two weeks at 20°C
  - Survives for seven to fifteen days at 37°C
  - Survives for three days at 50°C
- CSFV can survive for several months in refrigerated meat and several years in frozen meat.

Virus inactivation

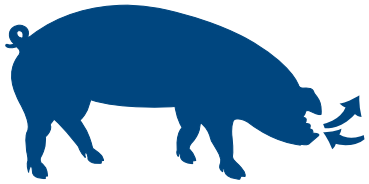
- pH sensitivity
  - Stable at pH 5-10
  - Inactivated at pH < 3 or pH > 11
- Disinfectant sensitivity
  - Organic solvents (ether, chloroform)
  - 2% sodium hydroxide
  - 1% formalin
- High temperatures
  - Heating at 65.5°C for over 30 minutes
  - Heating at 71°C for 1 minute

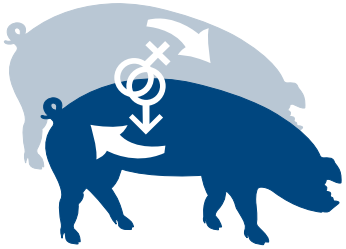
Transmission



- **Oral transmission**: Pigs ingest infected food or lick contaminated environments.
- **Direct transmission**: Contact with infected pigs, where the virus can enter through wounds or mucous membranes (such as eyes, nose, or mouth).
- **Environmental contamination**: Virus is shed into the environment through secretions, blood, saliva, urine, or feces of infected pigs, and can spread through contact with other animals or vehicles.

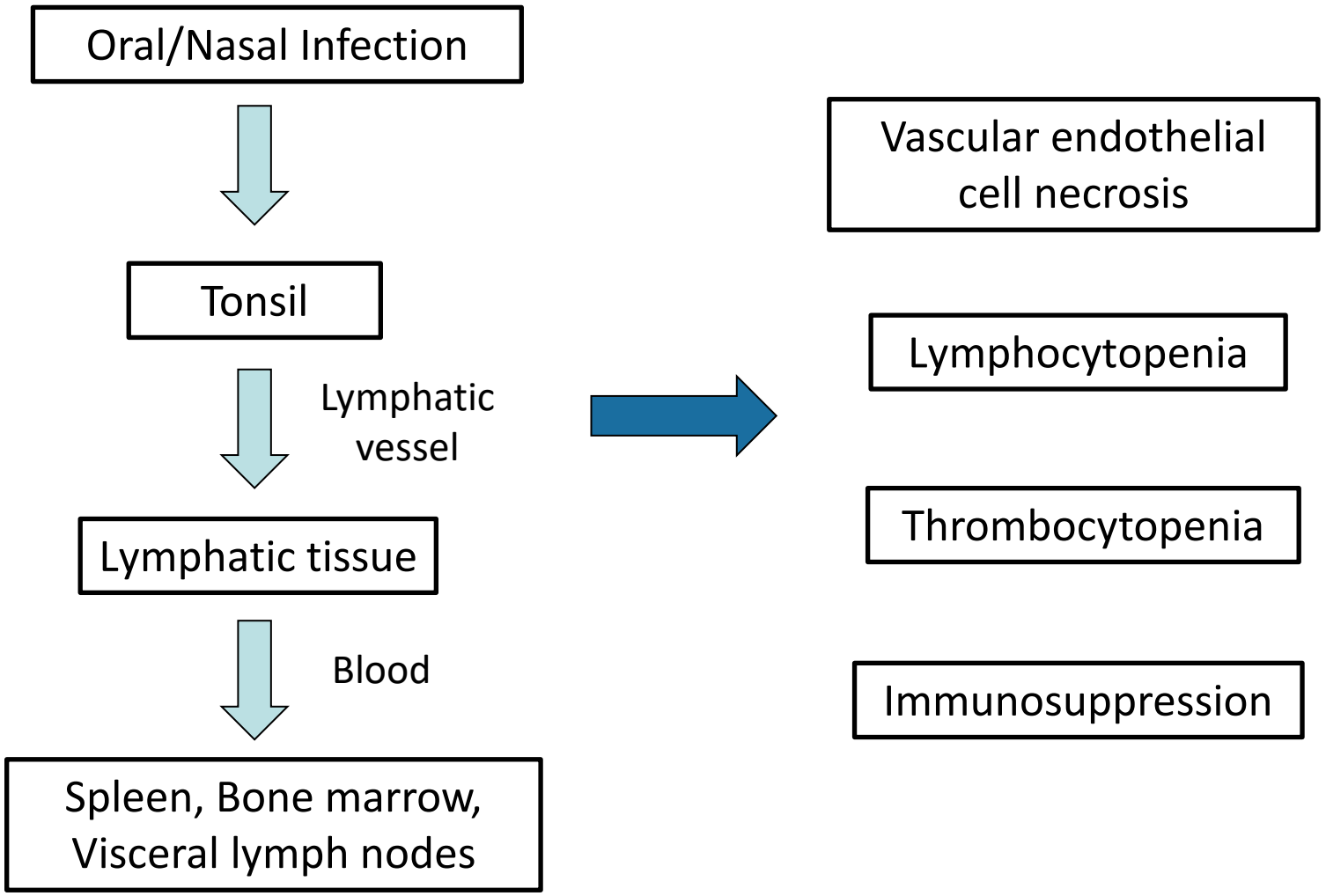
Transmission





- **Droplet transmission**: The virus can spread short distances through droplets.
- **Reproductive system transmission**: The virus can spread through semen during boar mating or infect fetuses during pregnancy in sows.

Pathogenesis



Clinical signs

- Acute form
- Chronic form
- Congenital form
- Postnatal form

Clinical signs – Acute form

- Highly virulent strains/infections in young pigs
  - High fever (> 41°C)
  - Depression
  - Cyanosis
  - Conjunctivitis
  - Constipation followed by diarrhea
  - Neurological symptoms
  - Abortion
  - High mortality rate

Clinical signs – Depression

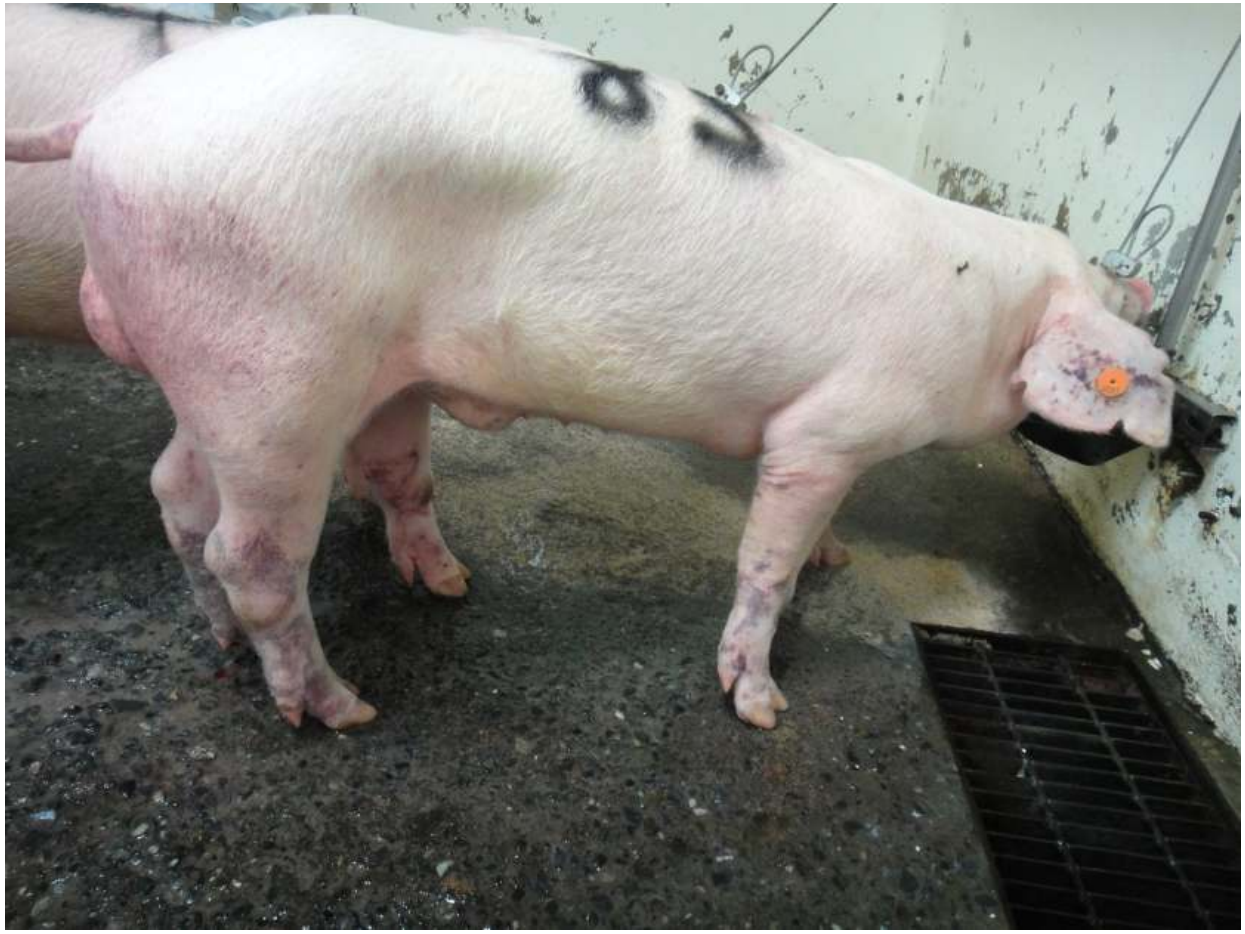


Clinical signs – Conjunctivitis





Clinical signs – Cyanosis



Clinical signs – Cyanosis





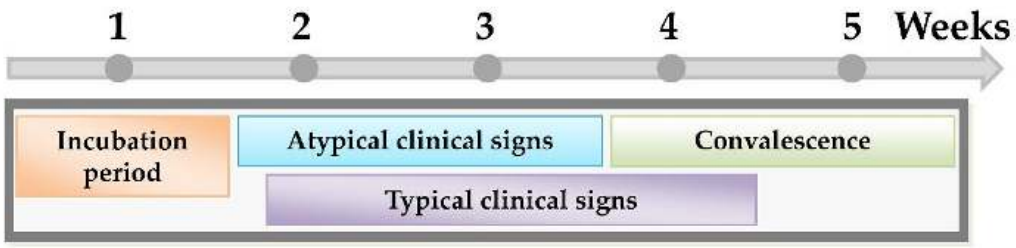
Clinical signs – Neurological symptoms



Clinical signs – Neurological symptoms



Clinical signs of acute classical swine fever infection

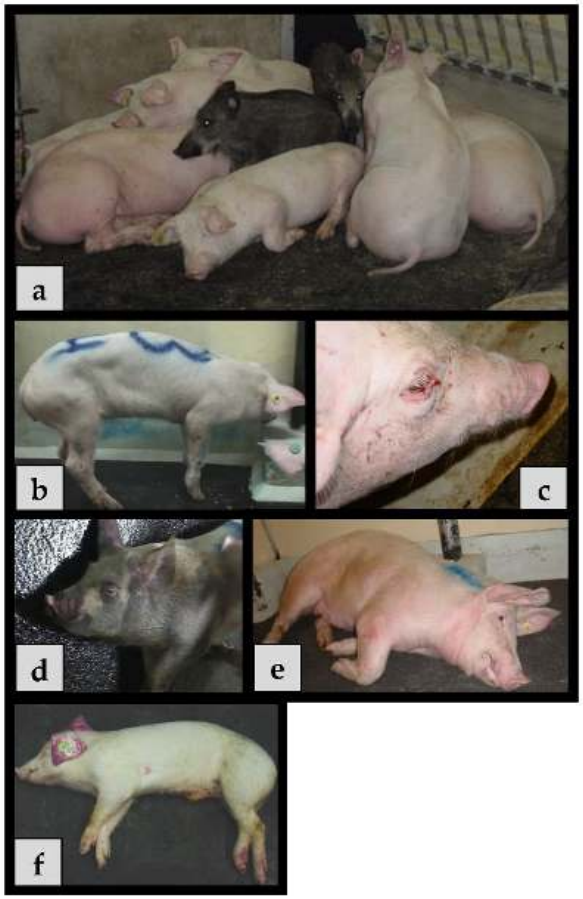


7–10 days: Incubation
Virus detection <7 days

1–3 weeks: Atypical clinical signs:
(High) fever
Inappetence
Depression
Conjunctivitis
Hunched back
Wasting
Respiratory signs
Gastrointestinal signs

2–4 weeks: Typical clinical signs:
Ataxia
Paresis and paralysis
Convulsions
Cyanosis
Petechiae
Ecchymoses

Secondary infections:
Pneumonia (purulent/fibrinous)
Enteritis



Blome *et al.*, 2017

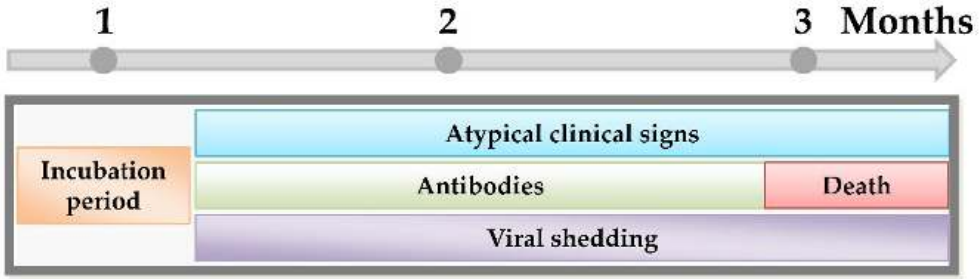
Clinical signs – Chronic form

- Low virulent strain/partially immune farms
  - Fever
  - Loss of appetite
  - Emaciation in pigs
  - Stunted growth
  - Death of pigs 2-3 months after infection

Clinical signs – Chronic form

- Infected pigs unable to generate an effective immune response against the virus.
- Infected pigs may appear healthy and asymptomatic.
- Infected pigs shed the virus throughout the clinical period until death.

Clinical signs of chronic classical swine fever infection

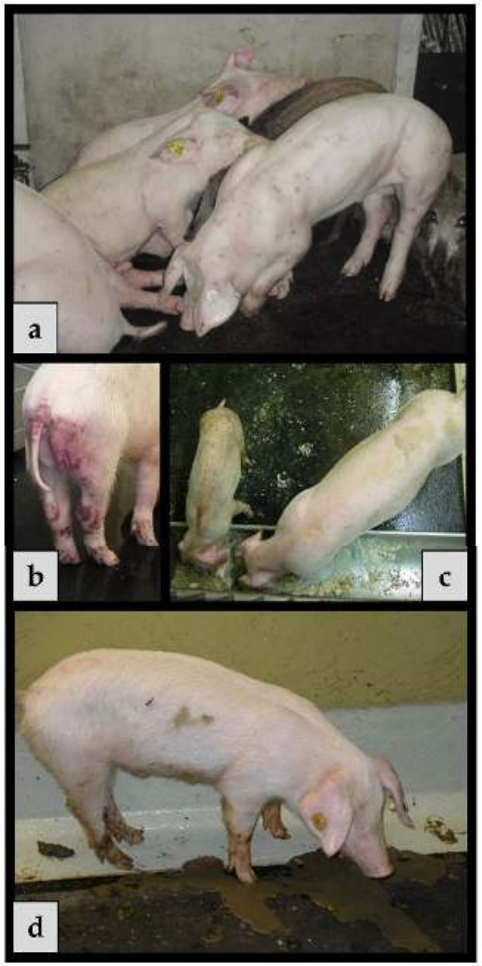


Up to 1 month: Incubation

1–3 months: Atypical signs
- Remittent fever
- Depression
- Stunted growth
- Wasting
- Diffuse dermatitis
- Respiratory signs
- Gastrointestinal signs

1–3 months: Viral shedding

Common secondary infections:
- Pneumonia (purulent/fibrinous)
- Enteritis



Blome *et al.*, 2017

Clinical signs – Congenital form

- Congenital persistent infection
- When a sow is infected with the virus, it can be transmitted to the fetus through the placenta.
- The course of the infection depends on the virulence of the virus strain and the timing of infection during pregnancy.
  - Stillbirths or the birth of weak, small piglets.
  - If the sow is infected during mid-pregnancy (between 50 to 70 days), the piglets born may have persistent viremia.

Clinical signs – Congenital form

- Late-onset CSF
- **Immunotolerance**: When the fetal immune system is immature, it cannot recognize the pathogen to produce antibodies against the virus.
  - Infected pigs may appear healthy or show atypical symptoms.
  - Infected pigs can survive for several months after infection but eventually die.
  - Infected pigs shed the virus throughout their life.
  - Antibodies may not be detected.
  - Vaccination effectiveness is poor.
  - The virus persists in the environment.

Clinical signs – Postnatal form

- Postnatal persistent infection
- Newborn piglets become infected with the virus.
- **Immunosuppression**: The piglets' immature immune systems are unable to produce antibodies against the virus.
  - Infected pigs may be clinically healthy or exhibit atypical symptoms.
  - Infected pigs can survive for several months after infection but eventually die.
  - High viral replication and shedding in the absence of adaptive immune response against the virus.

Gross lesions

- Peripheral hemorrhage of lymph node
- Infarction of spleen
- Petechial hemorrhage in kidney, bladder, and other organs

The pig died from the CSFV challenge

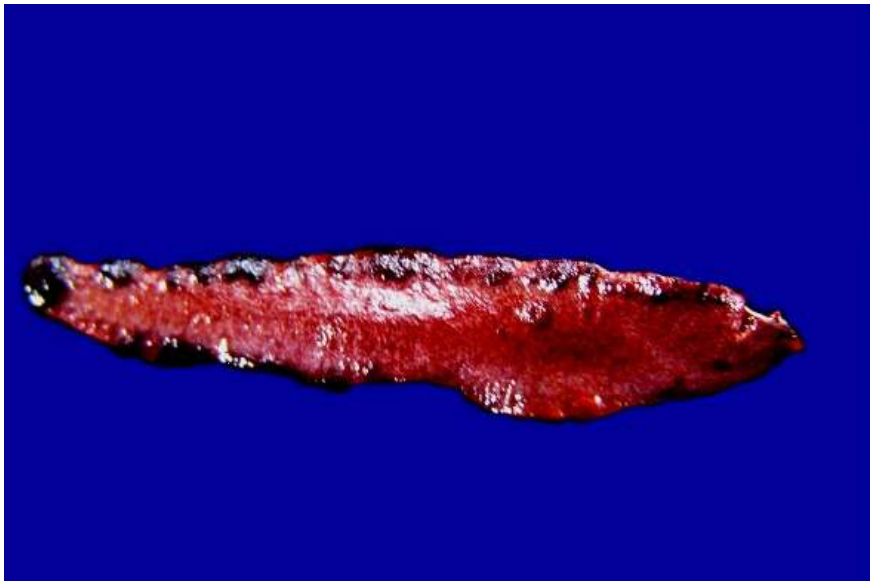


Peripheral hemorrhage of lymph nodes



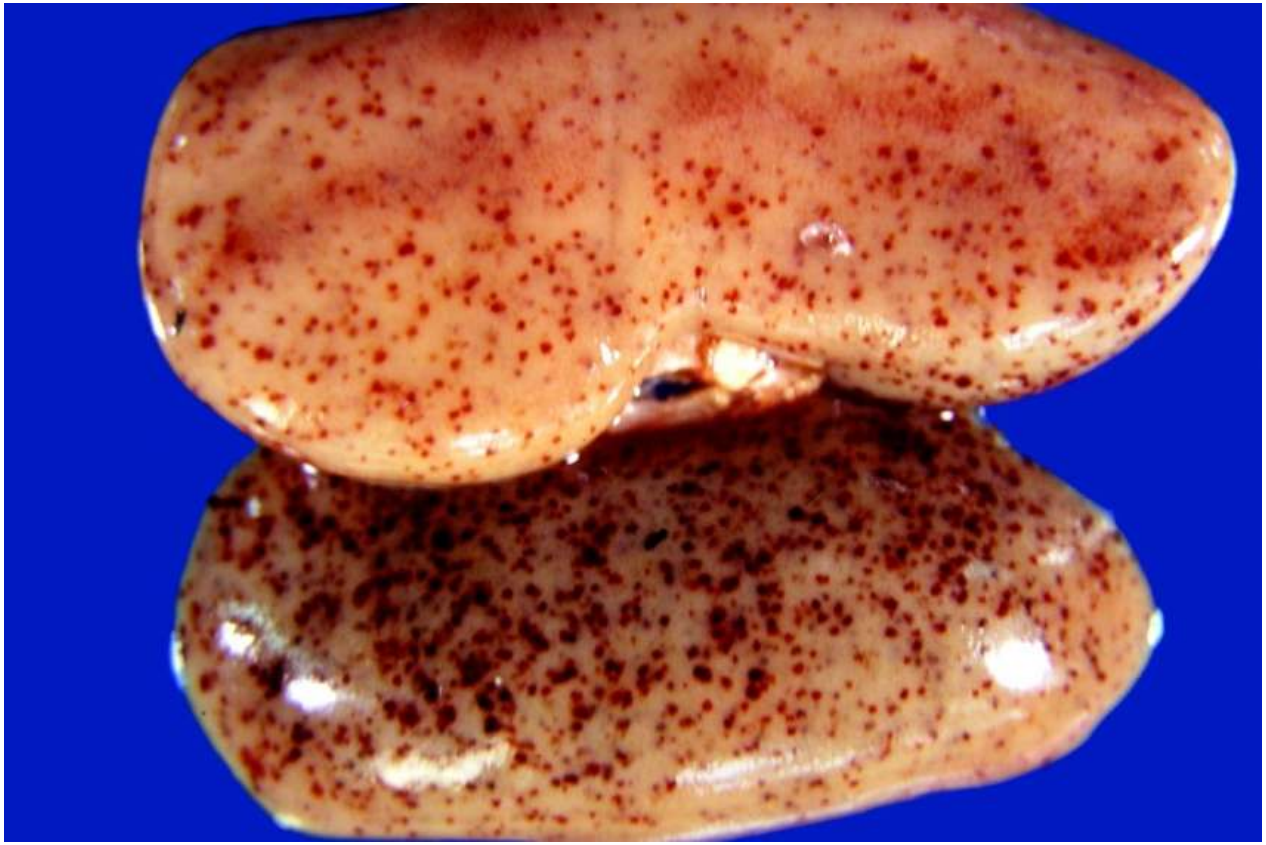


Infarction of spleen

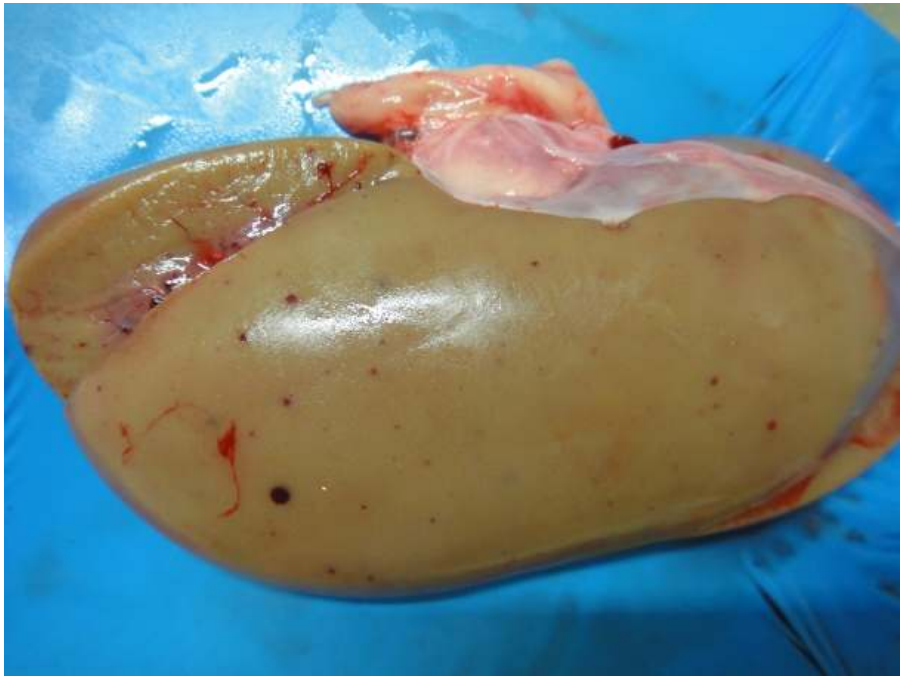




Petechial hemorrhage in kidneys

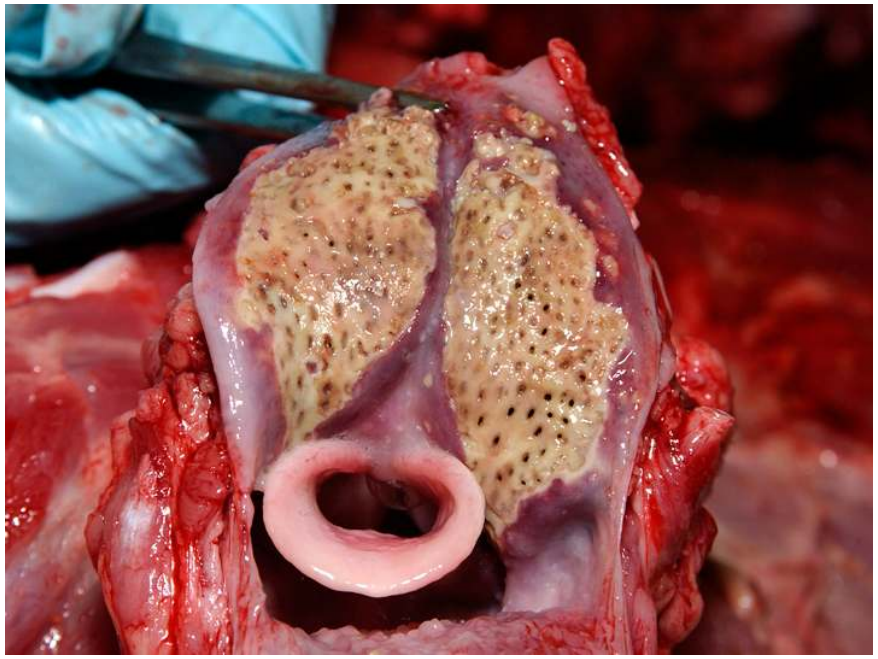


Petechial hemorrhage in kidneys





Necrosis of tonsil

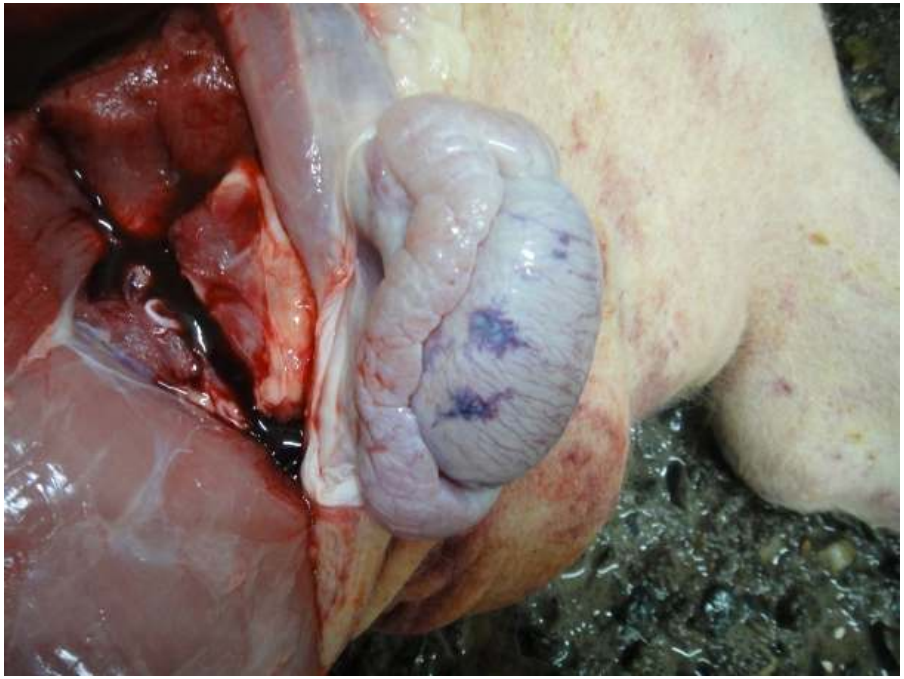




Petechial hemorrhage in bladder

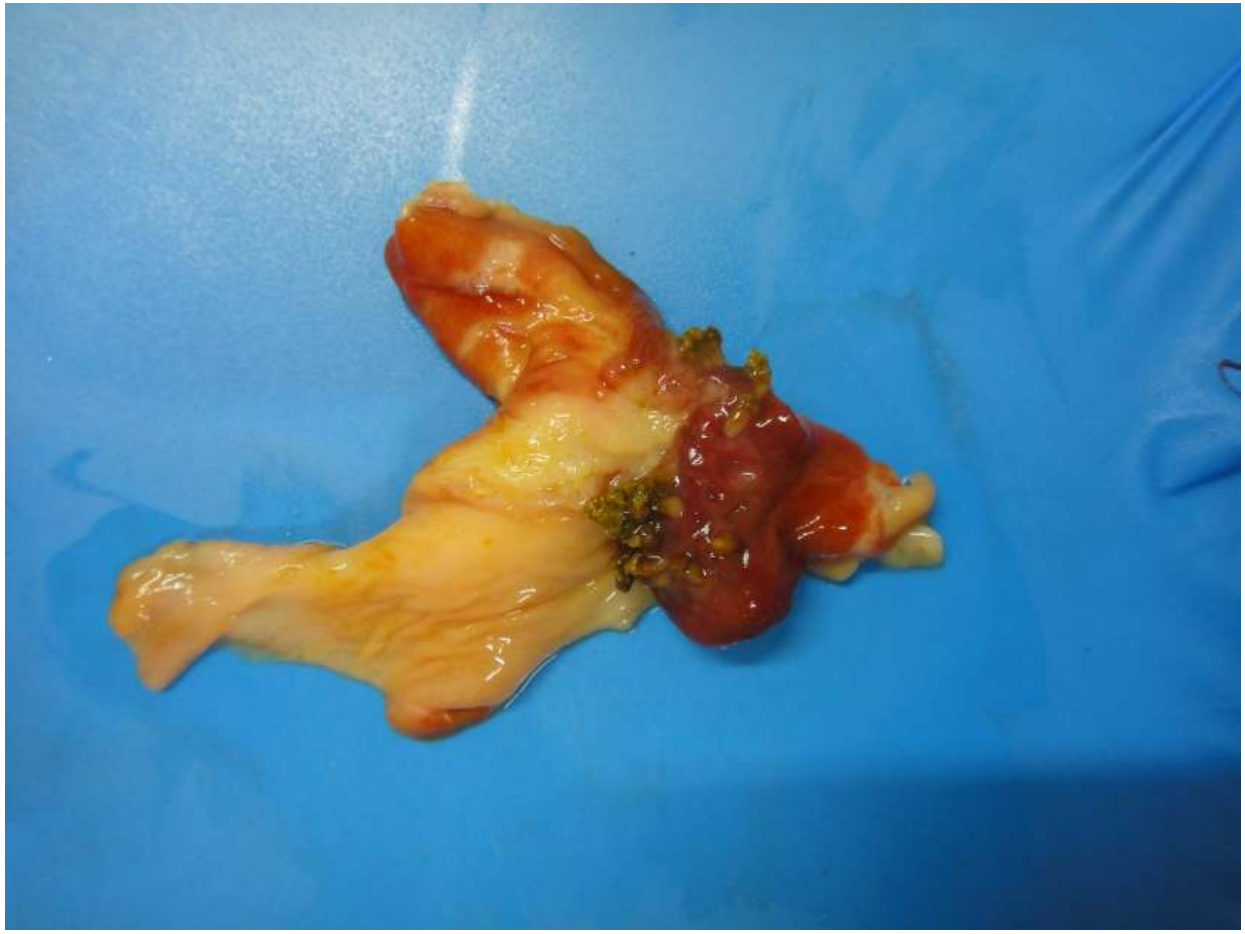


Petechial hemorrhage in testis





Ulcer of the ileocecal valve



Button ulcers in the mucosa of the gastrointestinal tract





Differential diagnosis

- African swine fever
- Porcine reproductive and respiratory syndrome (PRRS)
- Porcine circovirus-associated disease (PCVAD)
- Pseudorabies
- Septicemia caused by bacterial diseases
  - Erysipelas
  - Salmonellosis
- Trichinellosis

ASF vs CSF in Germany

- The clinical symptoms of ASFV intermediate virulent strains are no longer like the typical symptoms and lesions such as massive bloody diarrhea and spleen enlarged.
- Challenge experiments on pigs with ASFV or CSFV have shown that the clinical symptoms induced by the two infections are very similar.
- ASFV and CSFV never been isolated from the same pig.



Diagnosis – Antigen

- Virus antigen detection
  - Virus isolation: Cell culture
  - Virus nucleic acid detection: Nucleic acid amplification and sequence analysis

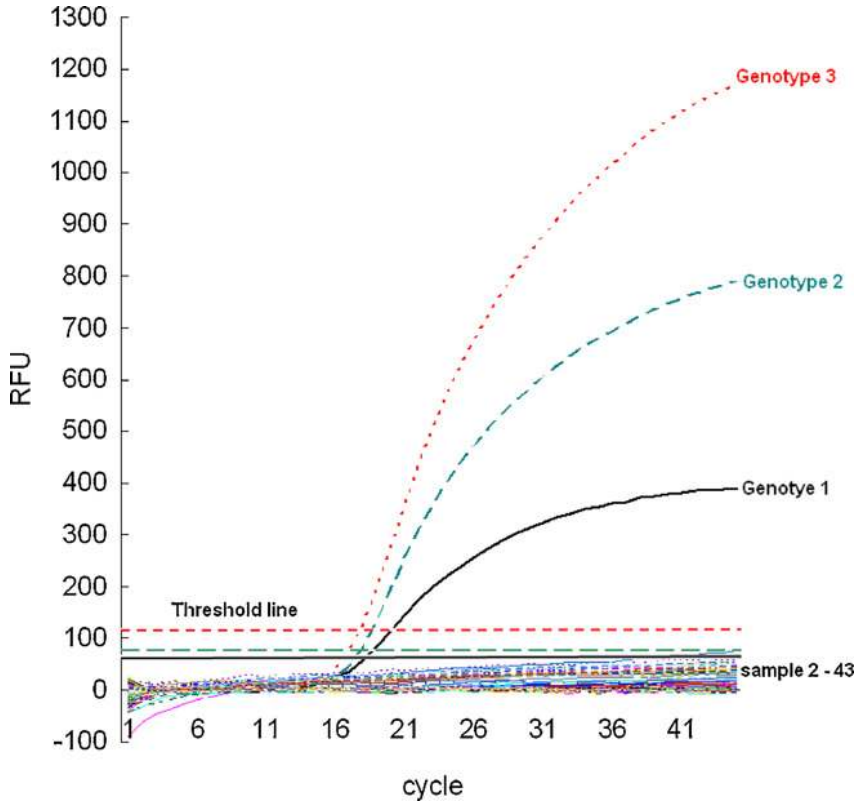
Diagnostic samples

- Tonsil
- Spleen
- Lymph nodes
- Kidneys
- Brain
- Ileocecal valve
- Blood

Virus nucleic acid detection

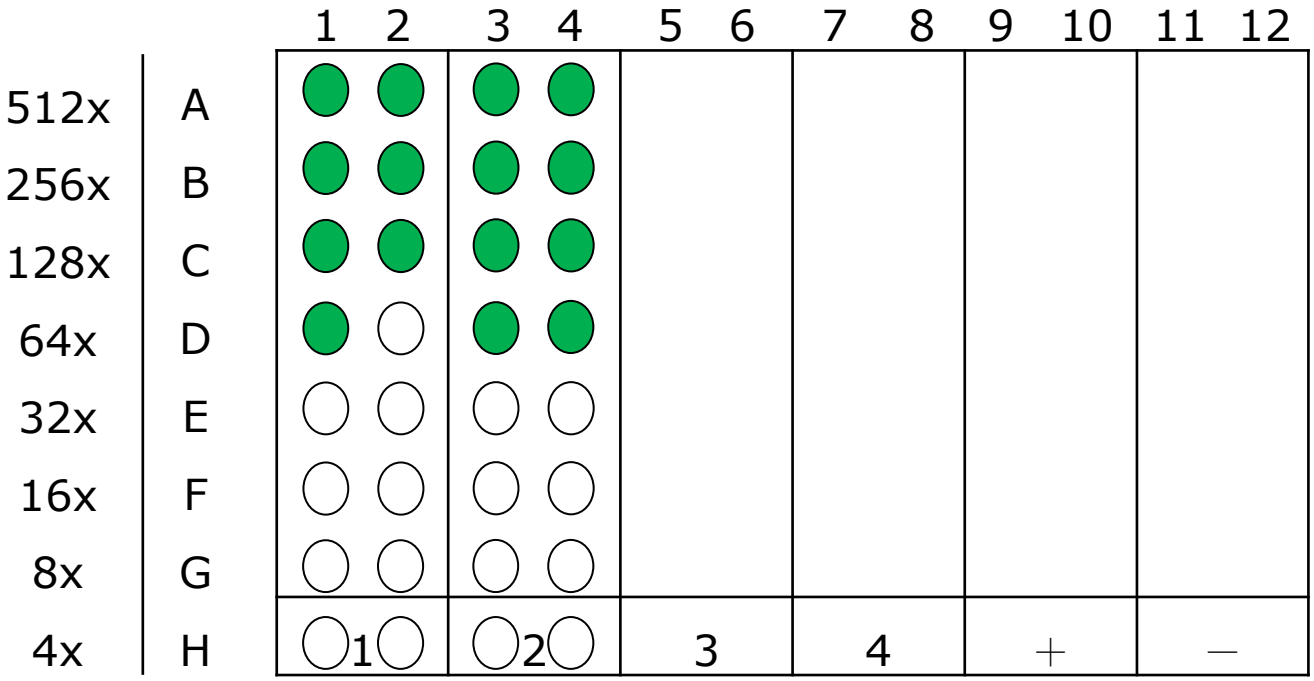
- RT-PCR
  - Nucleic acid amplification reaction + electrophoresis: Approximately 4 hour
  - Sequencing reaction: One day
- qRT-PCR
  - Distinguish different genotypes
  - Rapid response: Approximately 2 hours
  - High sensitivity and specificity

qRT-PCR for genotyping



- Distinguish between vaccine strain and field strains of CSFV

Antibody virus neutralization test



- The neutralizing antibody titer is determined as the highest dilution factor of serum that results in 50% fluorescence inhibition.
- Serum samples 1 and 2 have a neutralizing antibody titer of ___-fold.

ELISA

- Evaluating herd immunity status
- Commercial kits available



Diagnosis – Antibody

- FAVNT
  - Gold Standard
  - Laborious testing steps: Requires cell culture and virus handling
  - Time-consuming: Three days
- ELISA
  - Simple testing steps: No need for cell culture or virus handling
  - Quick response: Half a day
  - Requires evaluation of correlation with neutralizing antibodies

Atypical porcine pestivirus (APPV)

- APPV belongs to the genus *Orthopestivirus* of the family *Pestiviridae*.
- APPV is associated with congenital tremor type A-II in newborn piglets.



BVDV infected pigs



ge: www.elsevier.com/locate/vetmic

BVDV infected pigs

- Infected pigs typically show no clinical symptoms.
- Infected pigs may have no lesions or mild lesions.
- The infection source is usually in contact with cattle or other ruminant animals.
- Contamination of swine vaccine raw materials with BVDV.

BDV infected pigs



Cross-reactivity among pestiviruses of pig sera

Table 2. Cross-reactivity of sera taken from the individual animals at the end of the experiment. The samples were analyzed by a CSFV E2 antibody ELISA and by NTs against the indicated viruses.

Virus	Animal Number	CSFV ELISA	NT [4]				
			BVDV NADL	BDV Moredun	CSFV Alfort/187	CSFV Diepholz	Homologous Virus
TSV	3085	pos. [1]	20	20	-	160	640
TSV	3090	neg. [2]	-	-	-	40	240
TSV	3092	dbtf. [3]	15	10	-	160	480
TSV	3096	neg.	-	-	-	30	640
BDV	3087	neg.	10	30	-	240	1280
BDV	3088	neg.	-	20	10	320	1280
BDV	3094	neg.	40	30	20	480	1920
BuPV	3080	neg.	-	-	-	-	2560
BuPV	3086	neg.	-	-	-	-	6400
BuPV	3093	neg.	-	-	-	-	3200

Meyer *et al.*, 2021

Interpretation of neutralizing antibody

- How to determine if there is cross-reactivity in neutralizing antibody titers between CSFV and other viruses belonging to the genus *Pestivirus*?

Prevention and Control

- Vaccination
- Surveillance
- Biosecurity measures

Vaccination

- Live attenuated vaccine
  - Provides complete protection against field strains
- Subunit vaccine
  - Differentiate infected from vaccinated pigs (DIVA)



Prevention and Control

- In areas where the disease is **endemic**
  - Vaccination can prevent the spread of the disease.
  - DIVA vaccines are considered a feasible option for future control and elimination of CSF.
- In **disease-free** areas
  - A stamping-out policy includes early detection, movement control, cleaning, and disinfection.
  - This policy has eliminated CSF from North America and much of Western Europe.

CSF in wild boars

- Wild boars play a role in the transmission of CSF in parts of Europe and Japan.
- In Japan, the migration of wild boars makes the control of CSF difficult.



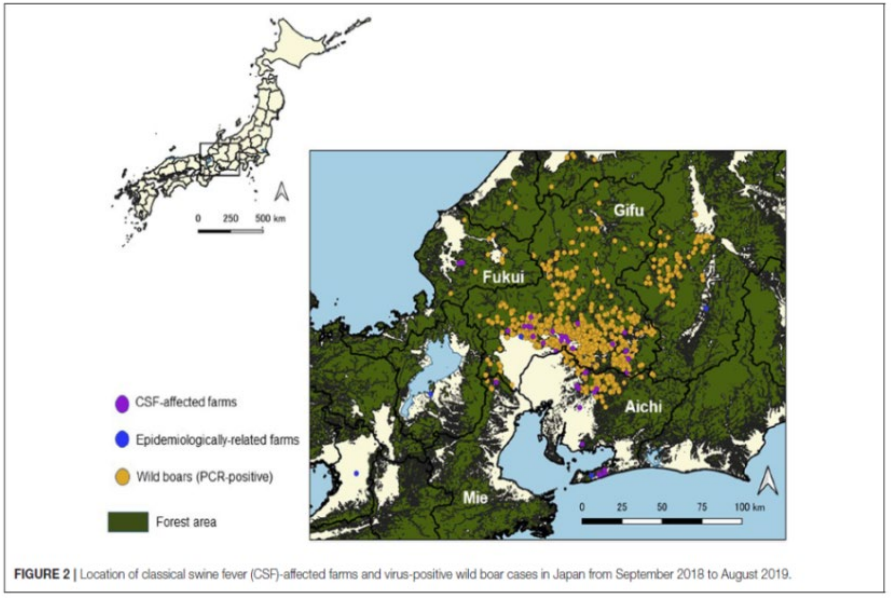


FIGURE 2 | Location of classical swine fever (CSF)-affected farms and virus-positive wild boar cases in Japan from September 2018 to August 2019.

Shimizu et al., 2020

CSF

岐阜新聞　2019年（平成31年）3月25日 月曜日

野生イノシシを媒介とした豚コレラの感染拡大防止のため、愛知県小牧市の山中にワクチン入りの餌を埋設する作業員＝24日（愛知県提供）



豚コレラ 愛知、ワクチン埋設開始

野生イノシシ感染防止 岐阜県も25日から

愛知県は24日、野生イノシシを媒介とした豚（とん）コレラの感染拡大を防ぐため、県内の山林でワクチン入りの餌の埋設を始めた。投与は国内初。県は効果を検証しながら、来年2月にか





岐阜県は25
する。
昨年9月に

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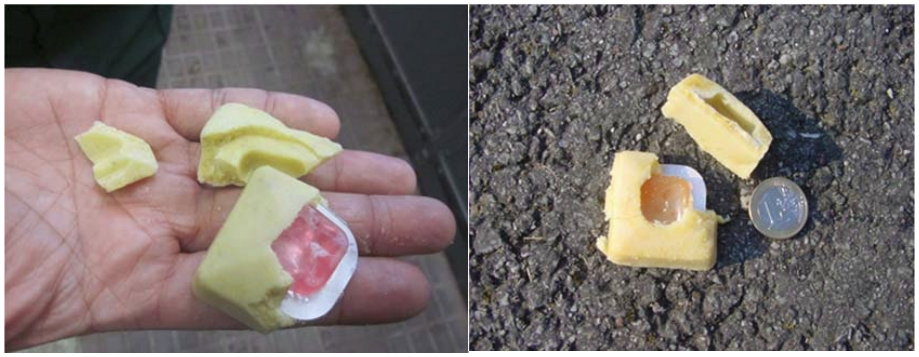
ゴルフの
2019（
阜放送主催
原市須衛天
で、プロと
チャリティ
プロから
2016年
選手をはじ
手ら男女12
会長、金森
原市長ら36
ホールア
クションを
競りに出さ
岐阜新聞岐
内の社会福

Oral vaccine in wild boars

(Reference) About the oral Vaccine

- ◯ Destination country: Germany
- ◯ size: 4 cm × 4 cm × 1.5 cm
- ◯ Composition:

  Outside: corn flour, paraffin, milk powder, almond flavoring, coconut oil
  Inside (Aluminum capsule): 1.6 ml liquid vaccine



Construction and size of the vaccine: Outside is Corn, and red liquid is the vaccine.



Hide bait vaccines in the ground or under stones, logs, etc... Then scatter normal feed around.

NIAH, Japan

CSF

月20日 金曜日　岐　阜　新　聞　明治25年3月31日第3種郵便物認可　第47804号　（日刊）

豚にワクチン接種

豚コレラ、国が方針転換

農水省 発生地中心に検討

農林水産省は19日、豚コレラ＝？参照＝対策として養豚場の豚へのワクチン接種を実施する方針を固めた。江藤拓農相が20日に、感染拡大を防ぐワクチン接種を可能にするよう防疫指針の改定作業に着手すると表明する。ワクチン接種を実施する地域については豚コレラの発生県を中心に、農水省が有識者の会議などの議論も踏まえて検討する方向だ。

農水省はこれまでワクチン接種には慎重な姿勢を続けてきたが、方針を大きく転換する。ただ実施に伴う課題も多く、根絶に向けては多角的な対策が必要になりそうだ。

ワクチン接種に向けては、まず農相が防疫指針の改定を有識者会議に諮問。接種する地域なども含めて議論し、農相に答申・報告する。国民への意見公募を経て指針を改定後、国と都道府県で協議した上で防疫担当職員を養豚場に派遣して接種する流れになる見しだ。通常なら数カ月かるが、農水省はできるだ早期に実施できるよう手きを急ぐ方針。

豚コレラは昨年9月に内では26年ぶりに感染が認され、これまで岐阜、知、三重、福井、埼玉、野の6県の養豚場や畜産験場で発生。農水省はウルスを媒介する野生イノシ対策を強化して封じ込を図ってきたが、感染拡道府県知事などから早期のワクチン接種を求める声が上がっていることを踏まえ、農水省は発生県などに地域を限定してワクチンをワクチン域外へ出るたるこ題も今後まま た場ている間が出かからない。には

【関連記事7面に】

ワクチン接種までに想定される流れ

農相が防疫指針の改定を有識者会議に諮問

↓

接種地域なども含め検討し、農相に答申・報告

↓

国民からの意見公募を経て指針を改定

↓

国と都道府県が協議しワクチン接種方針を確定

↓

都道府県が防疫担当職員を



2019年（令和元年）
9月20日
金曜日

岐阜新聞

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岐阜新聞Web
www.gifu-np.co.jp

彼岸入り

きょうの予定

▷県内
ふれあいいけ花展（～21日、岐阜市）

▷全国
中部国際空港でＬＣＣ向け新旅客ターミナルビルが開業

社会医療法人 緑西厚生会
松波総合病院
まつなみ健康増進クリニック
●人間ドック・健診
●まつなみリサーチパーク（医学研究所）
羽島郡笠松町田代185-1
TEL.058-388-0111㈹
http://www.matsunami-hsp.or.jp

県内 咲き誇る「芸術の秋」 19面



県華道連盟加盟の13流派による「ふれあいいけ花展」がきょう岐阜市で開幕。「秋らしい彩りや季節感を味わってほしい」

スポーツ 安打数ミスター超え 20面

プロ野球阪神の近本が今

CSF DIVA vaccine authorized by the EU

- Chimeric live vaccine (CP7_E2alf)
  - Suvaxyn CSF Marker, Zoetis
  - CSFV E^{rns} ELISA
- E2 subunit vaccine
  - Porcilis Pesti, MSD Animal Health
  - CSFV E^{rns} ELISA

Chimeric live vaccine

- Chimeric live vaccine (CP7_E2alf)
  - BVDV backbone with E2 gene replaced by CSFV E2
  - CP7_E2alf offers advantages as a live attenuated vaccine and can differentiate infected from vaccinated animals
  - CSFV E^{rns} ELISA
  - Emergency vaccine bank

CSFV E^{rns} ELISA

- CSFV E^{rns} ELISA has poor sensitivity and specificity.
- Antibodies to BVDV and BDV E^{rns} may cross-react with CSFV E^{rns} ELISA, causing false-positive results.
- Only suitable for population screening



Surveillance

- Comprehensive monitoring
  - Antibody testing: Assessing vaccine-induced herd immunity
  - Antigen testing: Monitoring the epidemic

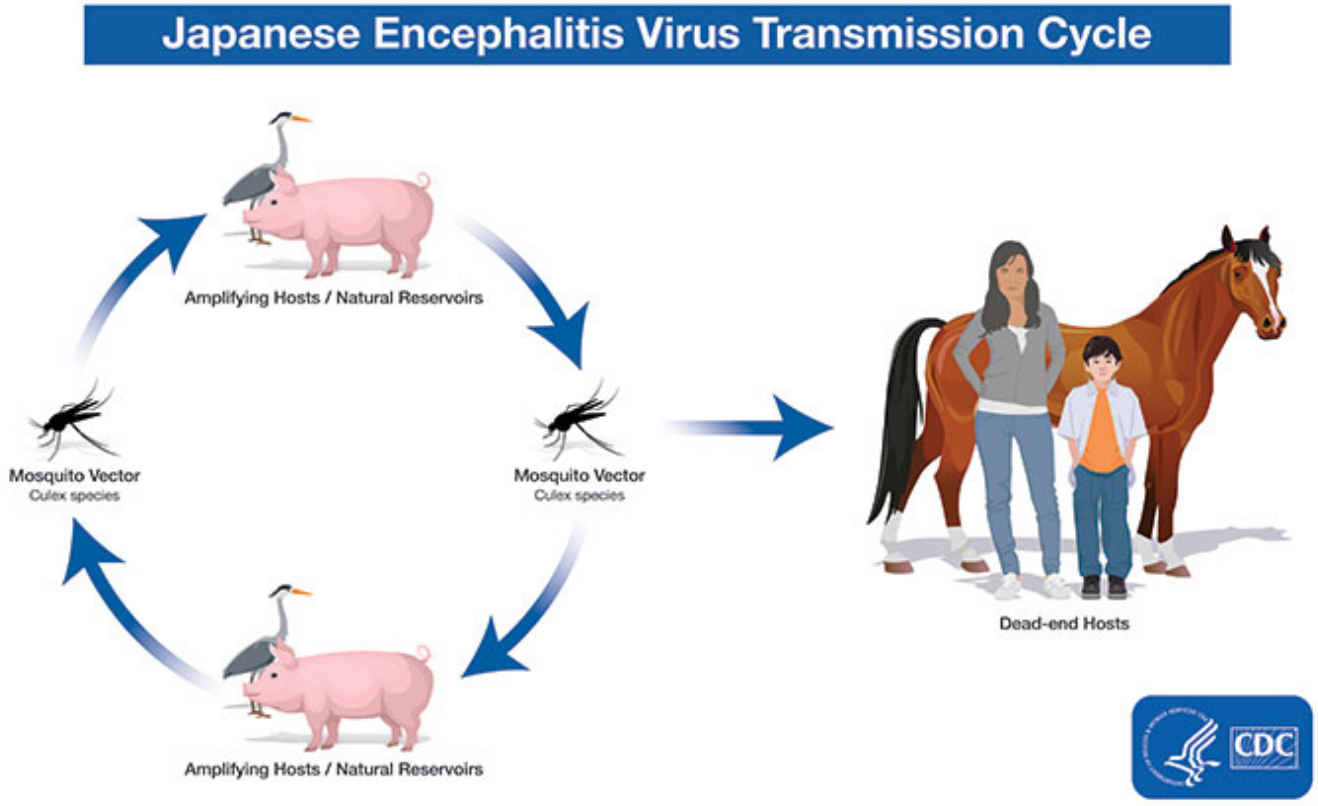
CSFV in Taiwan

- Virus populations have switched from **3.4** to **2.1** strains in the field after 1996.
- Both genotype **2.1** and **3.4** strains are moderately virulent.
- LPC vaccine is protective against both genotype **2.1** and **3.4** strains.
- No virus was detected in the field after 2006.
- Stop vaccination since 2023.

Take-home messages

- CSF is a highly contagious, non-zoonotic viral disease of pigs with a wide range of clinical outcomes.
- Effective vaccines are available and can provide rapid protection, but vaccination can complicate surveillance and disease elimination.
- Successful eradication requires a vaccination strategy, surveillance, diagnosis, movement control, and a carefully planned cessation of vaccination.
- Taiwan demonstrates that long-term surveillance and stepwise vaccination cessation can lead to successful disease elimination.

Japanese encephalitis

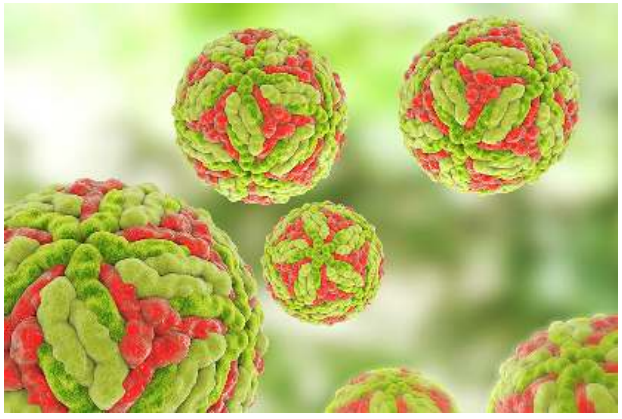


Japanese encephalitis virus (JEV)

- JEV was identified as the etiological agent in 1933 in Japan and subsequently isolated from *Culex tritaeniorhynchus* mosquitoes.
- JEV is a major cause of human viral encephalitis.
- JEV can also cause fatal encephalitis in horses.
- In pigs, infection with JEV can result in reproductive failure and can occasionally cause neurological disease in piglets.

Japanese encephalitis virus (JEV)

- JEV belongs to the family Flaviviridae and the genus *Orthoflavivirus* (formerly *Flavivirus*).
- The flavivirus genome is a single-stranded positive-sense RNA molecule approximately 11 kb in length.
- JEV is a mosquito-borne flavivirus, and belongs to the same genus as dengue, yellow fever, and West Nile viruses.



Kateryna Kon/science Photo Library

Virus inactivation

- Flaviviruses can be inactivated with various chemicals and disinfectants
  - 3mM binary ethylenimine, 3–8% paraformaldehyde, 2% glutaraldehyde, 1% hypochlorite, ethanol, isopropanol, and 0.05% Tween 20
- The pH is also important for viability.
- The thermal inactivation point of flaviviruses is 40°C, and the virus present in serum or media can be inactivated by heating at 56°C for 30 minutes.

Public health

- An estimated 100,000 clinical cases of JE occur globally each year, with more than 3 billion people living in areas at risk.
- Approximately 75% occur in children under 15 years of age.
- Amplification of JEV in pigs is generally thought to precede epidemic transmission.
- Epidemic activity in human populations can occur in the absence of large populations or high densities of pigs.

Epidemiology

- JEV is a zoonotic virus maintained in nature by transmission cycles involving *Culex* sp.
- **Mosquitoes** and certain species of wild and domestic birds and **pigs** as the vertebrate hosts.
- **Humans** and **horses** become infected when bitten by an infected mosquito, but they are incidental, **dead-end hosts**.

Epidemiology

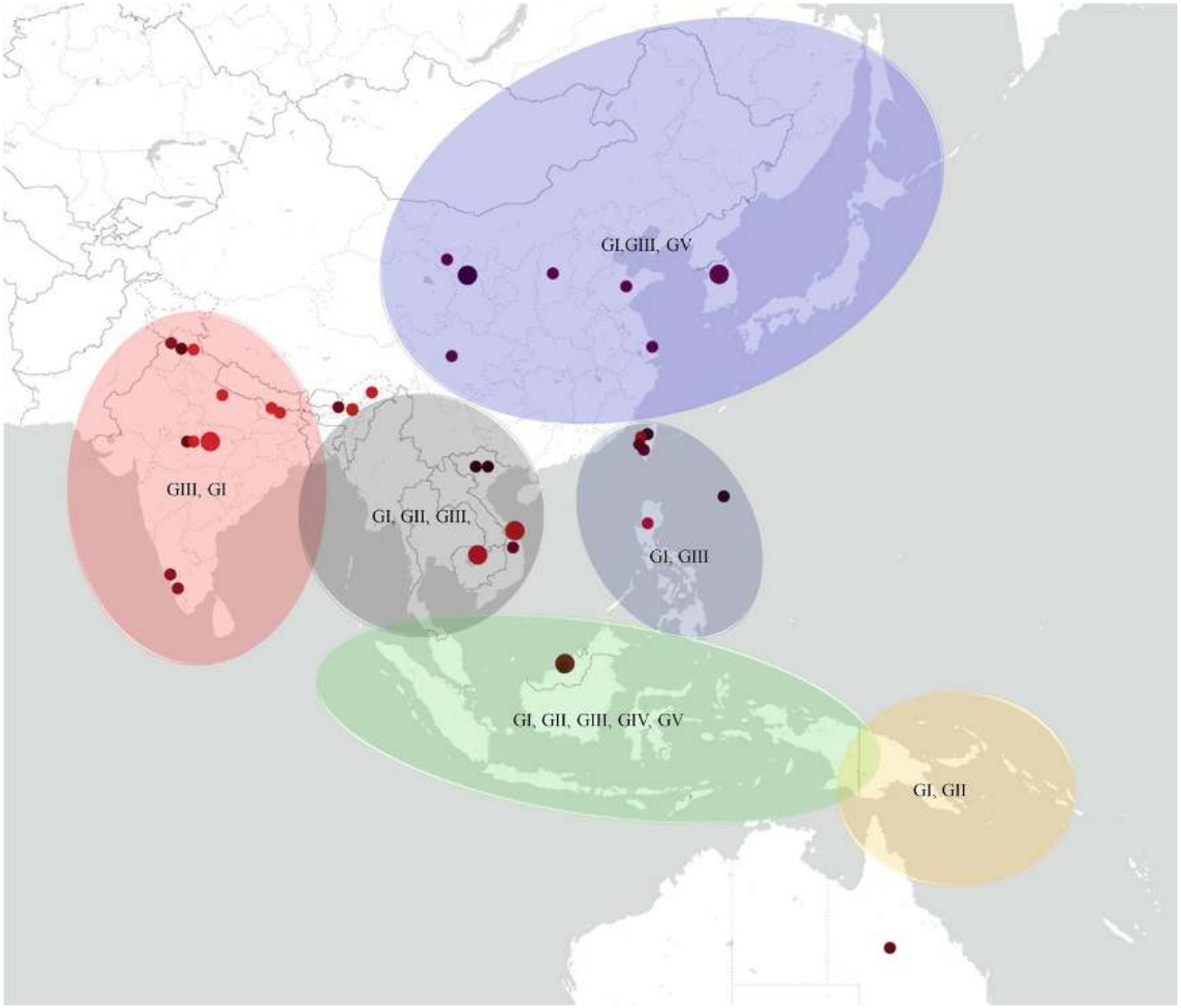


Source: CDC

Genotypes of JEV

- JEV exists as a single serotype.
- There are five genotypes of JEV
  - **Genotype I** comprises strains from Cambodia, China, Korea, northern Thailand, Vietnam, Japan, India, Australia, and **Taiwan**.
  - Genotype II comprises strains from Indonesia, Malaysia, southern Thailand, Papua New Guinea, and Australia.
  - **Genotype III** is widely distributed among Asian countries such as Japan, China, South Korea, and **Taiwan**.
  - Genotype IV isolates have only been found in Indonesia.
  - Genotype V strains have been found in Malaysia, China, and South Korea.
  - In the past decades, **genotype I** has replaced circulating genotype II and genotype III viruses in several regions.

Genotypes of JEV

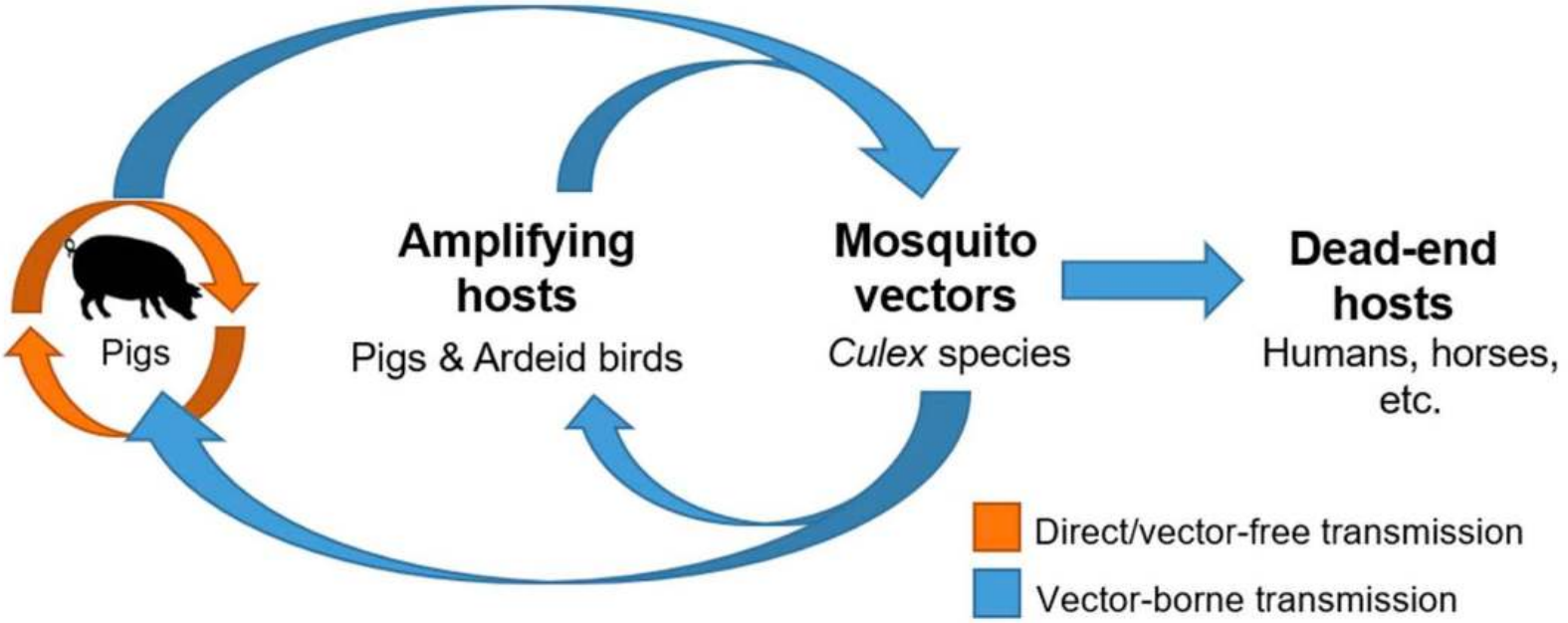


Saxena et al., 2020

Transmission

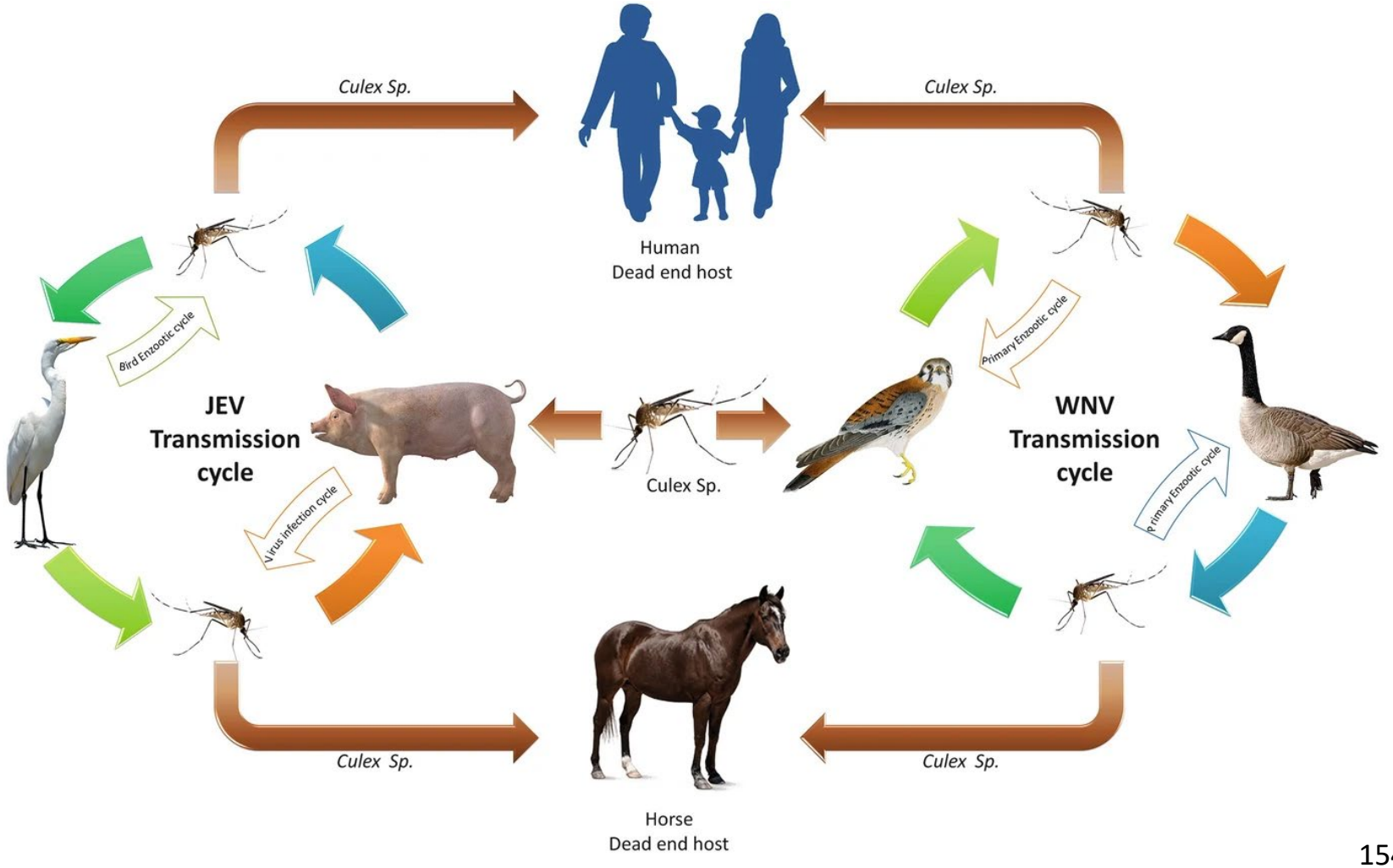
- **Pigs** are the principal **amplification hosts** of JEV and play an important role in virus amplification in endemic and epidemic areas.
- Pigs develop high-titer viremia following infection, typically lasting 2–4 days and sufficient to infect mosquito vectors.
- Pigs consistently display higher virus titers than other domestic or wild animals, as a sensitive indicator of virus transmission in endemic areas.
- **Wild birds**, particularly ardeid species, are believed to be important maintenance hosts of JEV and may act as amplifiers in epidemics.

Transmission



Park et al., 2022

Transmission of JEV and WNV



Saxena et al., 2020

Pathogenesis

- Flavivirus infections are initiated following the bite of an infected **mosquito**.
- Contact transmission between pigs may also play a significant role.
- The virus can reach the central nervous system (CNS) as early as 3 DPI.
- Neonates and young animals appear to be more likely than adults to develop severe CNS infections.

Pathogenesis

- JEV infection in pigs usually results in subclinical infection with viremia.
- JEV in pigs may infect neurons, leading to meningoencephalitis with neurological signs.
- Reproductive failure is also a recognized outcome of infection in adult pigs.
- JEV crosses the placenta to infect fetuses, causing abortion, stillbirths, mummification, and neonatal deaths.
- JEV may also infect testicular tissues in boars, causing infertility.

Clinical signs

- The most common disease manifestation in infected pregnant sows or gilts is **reproductive failure** leading to abortion and litters that contain stillborn or mummified fetuses or live, weak piglets.
- Reproductive failure occurs in sows infected with JEV before 60–70 days of gestation.
- Infections later in gestation generally have little or no effect on the fetuses.

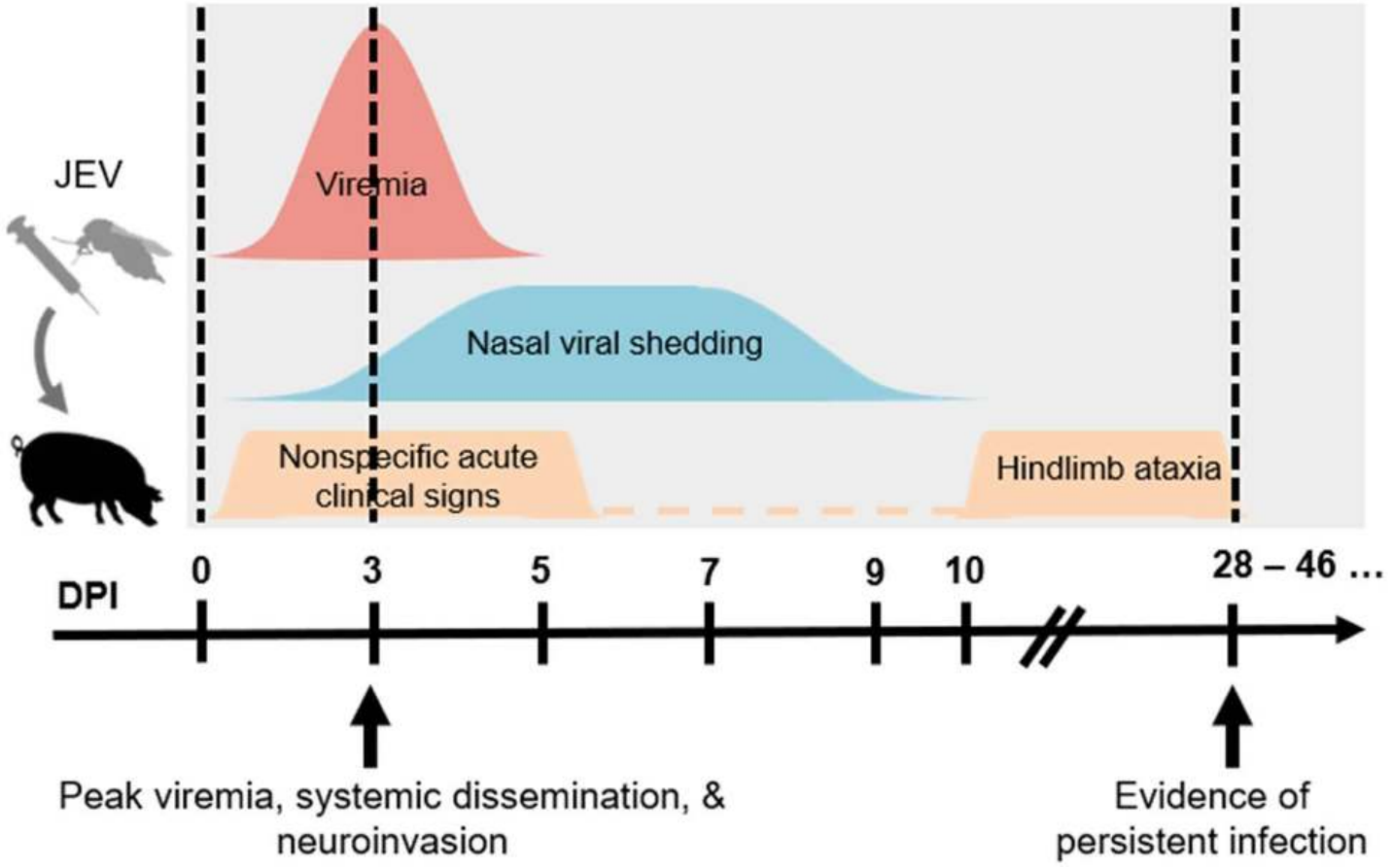


Diseases of Swine, 12th Edition

Clinical signs

- JEV has also been associated with infertility in boars.
- Infection of boars may lead to edematous, congested testicles, resulting in lower motile sperm counts and abnormal spermatozoa.
- These effects are usually temporary, and complete recovery occurs in most instances.

Infection outcomes of JEV in pigs



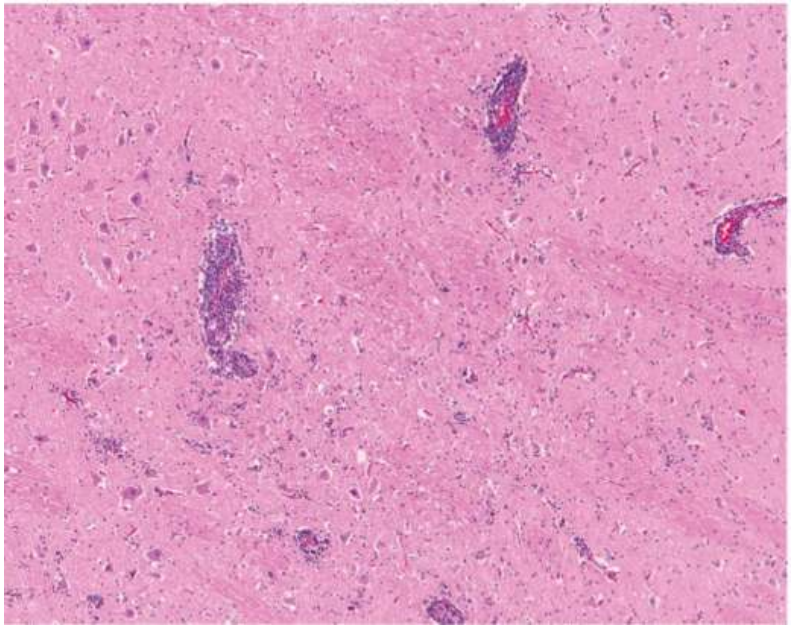
Gross lesions

- There are no characteristic lesions of JEV infection in sows that manifest as reproductive failure.
- In boars, there is a thickening of the tunica vaginalis and epididymis, with edema and inflammation of the epididymis, tunica vaginalis, and testis.
- Stillborn fetuses or live, weak piglets often present with hydrocephalus, subcutaneous edema, cerebellar hypoplasia, and spinal hypomyelinogenesis.

Microscopic lesions

- Infected piglets may present with diffuse nonsuppurative encephalitis, characterized by neuronal necrosis and perivascular cuffing in the brain and spinal cord.

Nonsuppurative encephalitis



Diseases of Swine, 11th Edition

Differential diagnosis

- Agents causing clinical diseases characterized by abortion, fetal mummification or stillbirth, and encephalitis in pigs up to 6 months of age
  - Pseudorabies
  - Classical swine fever virus
  - Porcine reproductive and respiratory syndrome (PRRS)
  - Hemagglutinating encephalomyelitis virus
  - Rubulavirus (blue eye paramyxovirus)
  - Menangle virus
  - Porcine teschovirus
  - Porcine parvovirus
  - Brucellosis
  - Salt poisoning

Diagnostic samples

- **Placenta and fetal tissues** – reproductive cases
- **Brain, spleen, liver, and tonsil** – virus/RNA detection
- **Whole blood** – virus detection during early viremia
- **Serum** – serology and early virus detection
- **Paired serum samples** collected **2–4 weeks apart** – demonstration of seroconversion

Diagnosis – Laboratory

- Antigen detection
  - Virus isolation
  - Reverse transcription polymerase chain reaction (RT-PCR)
  - Immunohistochemistry (IHC)
- Serological tests for antibody detection
  - ELISA
  - Immunofluorescent antibody (IFA)
  - Virus neutralization (VN) assays

Immunity

- Cross-neutralizing antibody responses to JEV have been observed in pigs following experimental infection with closely related flaviviruses.
- Similar booster responses occurred if infection with West Nile virus was followed by a challenge with JEV.
- In regions hyperendemic for flaviviruses, cross-reactive immunity in pigs from prior infection with related flaviviruses may be expected to inhibit subsequent infection with JEV.

Prevention and Control – Vaccination

- Inactivated, mouse brain-derived, or live attenuated vaccines have been used to prevent virus amplification or reproductive disease in pigs in Japan, Taiwan, Nepal, and Korea.
- The live attenuated vaccine is more efficacious than the inactivated vaccine against natural and experimental infection.
- The emergence of **genotype I** throughout much of Asia has raised concerns about the efficacy of **genotype III-based** vaccines.



Prevention and Control – Vaccination

viruses

MDPI

Article

Effectiveness of Live-Attenuated Genotype III Japanese Encephalitis Viral Vaccine against Circulating Genotype I Viruses in Swine

Yi-Chin Fan [1,2,†], Yi-Ying Chen [1,†], Jo-Mei Chen [1,†], Chienjin Huang [1], Mei Huang [3,*] and Shyan-Song Chiou [1,*]

Table 2. Neutralizing antibody against GI and GIII JEVs in serum samples collected from sows and gilts in 2016.

Positivity of Neutralizing Antibody (%)					
Against GIII JEV			Against GI JEV		
Sow	Gilt (Before Vaccination)	Gilt (After Vaccination)	Sow	Gilt (Before Vaccination)	Gilt (After Vaccination)
80	10	100	80	0	40
100	0	100	90	0	40
80	0	100	80	0	30
90	10	100	90	0	40
90	0	0	90	0	0
80	10	0	80	0	0

Fan et al., 2022

- The vaccine effectiveness of the **GIII** live-attenuated JEV vaccine to prevent stillbirth/abortion caused by **GI** viruses is lower than that against **GIII** viruses.

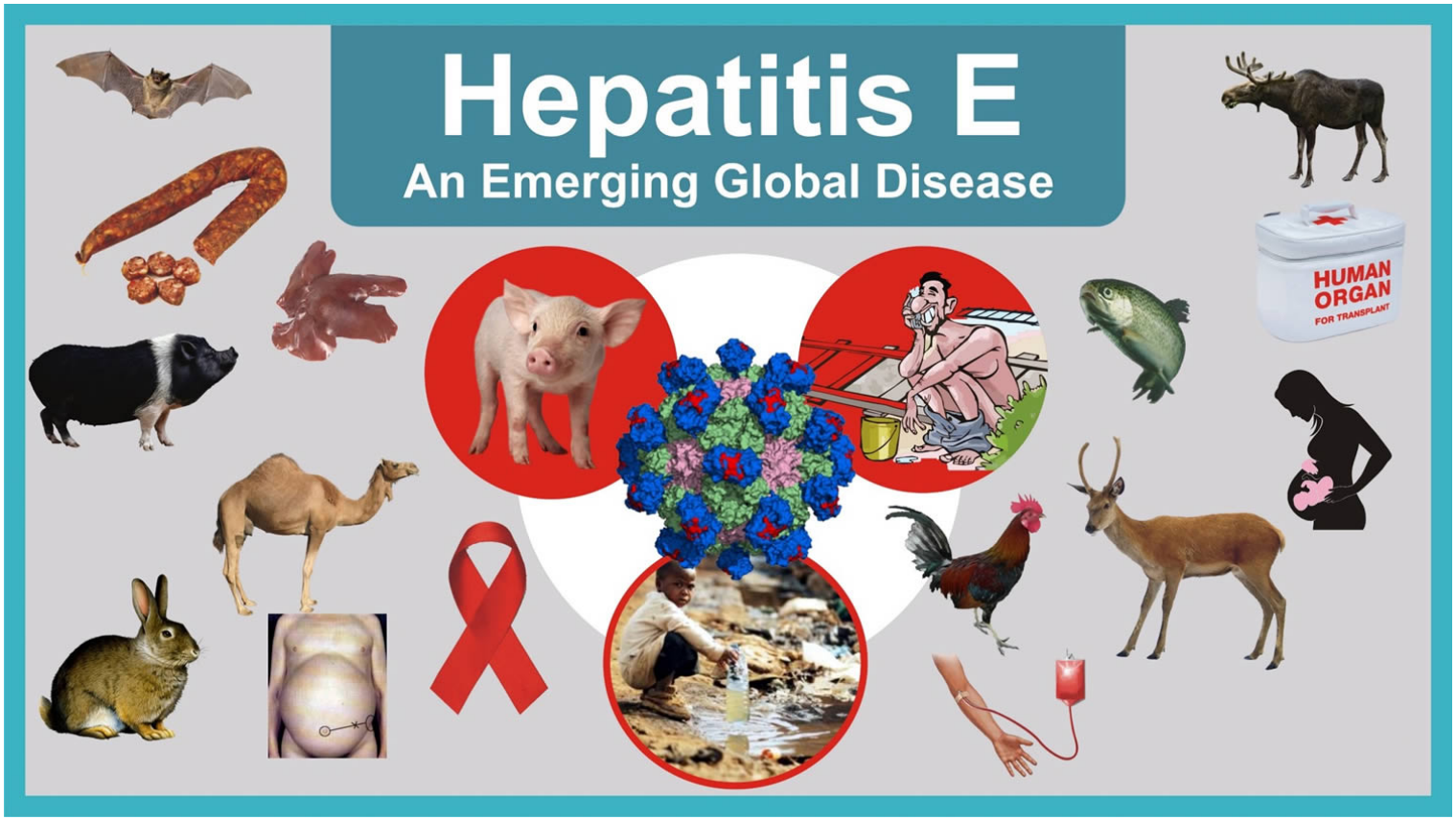
Prevention and Control

- JEV infection of pigs can be controlled by preventing their exposure to infected **mosquitoes**, but this is usually not practical unless pigs are maintained in a mosquito-free environment.
- **Mosquito** management and control programs implemented for public health purposes in the proximity of pig farming activities may also be expected to reduce mosquito-biting activity and therefore swine infections.

Take-home messages

- JEV is a mosquito-borne zoonotic virus, with pigs serving as major amplifying hosts.
- Most infections in pigs are subclinical, but JEV can cause reproductive failure and, occasionally, neurologic disease in young pigs.
- Humans and horses are dead-end hosts and do not contribute significantly to virus amplification.
- Vaccination and mosquito control are important preventive measures; genotype shifts may influence vaccine performance.

Hepatitis E

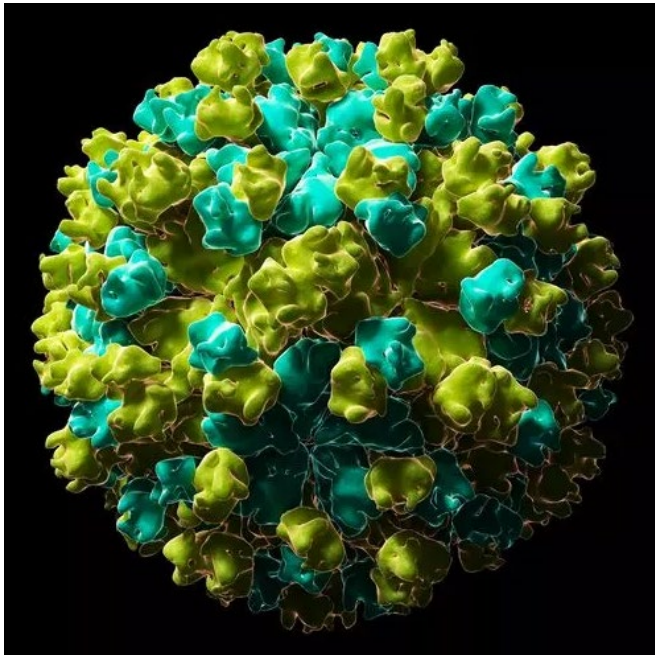

Hepatitis E
An Emerging Global Disease
HUMAN ORGAN
FOR TRANSPLANT

Hepatitis E virus infection

- Hepatitis E virus (HEV), the causative agent of human hepatitis E, is an important public health problem in many developing countries in Asia and Africa.
- Hepatitis E is also endemic in industrialized countries, including the United States and many European countries.
- Although swine HEV only causes microscopic lesions of hepatitis in pigs with **no** signs of clinical disease, it poses a **zoonotic risk** to humans via direct contact with infected pigs and consumption of undercooked pork or pork products.

Hepatitis E virus (HEV)

- HEV is classified in the family *Hepeviridae*.
- The genome of swine HEV is a single-stranded, positive-sense RNA molecule of approximately 7.2 kb.



Hepatitis E virus (HEV)

- 8 distinct genotypes (HEV-1 to HEV-8):
  - HEV-1 and HEV-2 are restricted to humans
  - **HEV-3** and **HEV-4** infect humans, pigs and other species
  - HEV-5 and HEV-6 infect wild boar
  - HEV-7 and HEV-8 infect camels

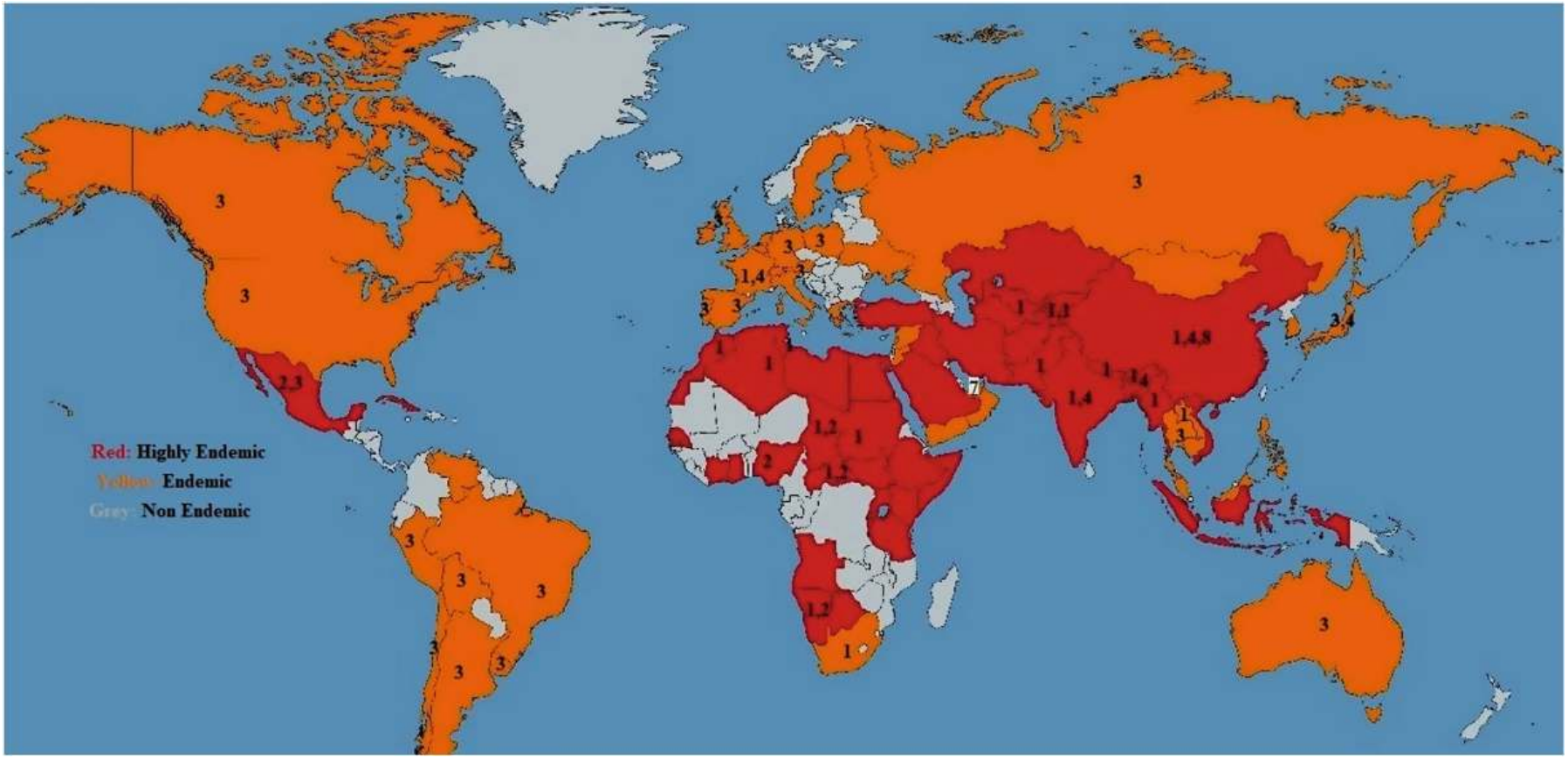
Virus inactivation

- HEV is thought to resist inactivation by acidic and mild alkaline conditions in the intestinal tract.
- The infectivity of swine HEV present in commercial pig livers is completely inactivated by adequate cooking.
- Incubation of the contaminated pig liver homogenates at 56°C for 1 hour did not abolish virus infectivity.

Public health

- Hepatitis E is a zoonotic disease for which **pigs** are an important **reservoir**.
- Concerns for zoonotic infections include occupational exposures to infected pigs and consumption of contaminated pork or pork products.
- **HEV-3** and **HEV-4** strains from swine are **zoonotic** and infect humans.
- The zoonotic potential for HEV-5 and HEV-6 strains from wild boar remains unknown.
- HEV-infected pigs excreted large amounts of HEV in feces, thus posing a concern for environmental safety.

Genotype distribution



Kumar et al., 2020

HEV and Public Health

Age (years)	Swine farmers		Health examination attendees	
	n/N	Rate (95% CI)	n/N	Rate (95% CI)
<20	0/7	0.0 (0.3–37)	0/22	0.0 (0.1–15)
21–30	1/22	4.5 (1–22)	0/46	0.0 (0–7.5)
31–40	6/33	18.2 (9–35)	3/76	4.0 (1–11)
41–50	14/43	32.6 (20–48)	8/73	11.0 (6–20)
51–60	17/35	48.6 (33–65)	11/51	21.6 (13–35)
>61	8/16	50.0 (28–72)	14/46	30.4 (19–45)
All	46/156	29.5 (23–37)	36/314	11.5 (8–15)

Lee et al., 2013

- The prevalence of serum antibodies to hepatitis E among the general public was 11.5%, and that of **pig farmers** was 29.5%, which is **3.5 times** the risk of hepatitis E infection compared with the general public.

scientific reports

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Hepatitis E virus infection in 6-month-old pigs in Taiwan

Ming-Hui Liao[1,2,8], Fang-Tzy Wu[3,8], Huimin Bai[4], Yen Hai Doan[7], Jyh-Yuan Yang[3], Naokazu Takeda[5], Masamichi Muramatsu[6] & Tian-Cheng Li[6]✉

Hepatitis E virus (HEV) is the causative agent of acute hepatitis E. Genotype 3 (G3) and 4 (G4) HEV have recently been identified in and isolated from swine as the main HEV genotypes worldwide. However, there is limited information on HEV infection status among pigs in Taiwan, especially pigs in the stage before transportation to the slaughterhouse. To determine the frequency of HEV infection among pigs in Taiwan, we detected and quantified HEV RNA contained in 295 fecal specimens collected from 6-month-old pigs bred in 30 pig farms located in 8 counties. We found that [redacted] of the fecal specimens were positive for HEV RNA by a quantitative real-time reverse transcription-polymerase chain reaction, and the copy number ranged from 2.3×10^3 to 2.08×10^7 copies/g. Amplification of a 338 bp sequence in ORF2 was achieved in 16 of 74 HEV RNA-positive samples, and their nucleotide sequences were determined. Two HEV sequences appeared to belong to subtype 3a of G3 and the remaining 14 HEV sequences belonged to subtype 4b of G4 (G4b). The entire genome sequence of two G4b HEVs was obtained by next-generation sequence analyses, and the phylogenetic analyses indicated that unique G4b HEVs were circulating in pig farms in Taiwan. In the present study, we found that both G3 and G4 HEVs were circulating in Taiwanese pig farms and G4b was the predominant subtype. In addition, the relatively high detection frequency of HEV RNA in the 6-month-old pigs indicated that Taiwanese pigs just before transportation to the slaughterhouse are at risk of carrying HEVs, and thus thorough cooking or heating of pork meat or organs is needed before consumption in Taiwan and possibly in other countries as well.

Liao et al., 2020

HEV and Food Safety

Pork product type	Virus detection				
	HEV RNA Detected	No of samples tested	Valid extraction efficiency	% HEV positive per sample type	Genome copies/g
Pork sausage (72 g)	1	59	49	2	0.02
Pork sausage (27 g)	1	67	65	1.5	1.02
Fermented sausage	1	49	49	2	0.4
Pork liver	6	25	25	24	(0.26–9.4)
Total	9	200	188		

- The highest incidence of HEV RNA was in **pork liver**, where **24%** of samples were positive.

Bennett et al., 2024

Epidemiology

- Swine HEV infection is ubiquitous in pigs worldwide, regardless of whether HEV is endemic in the respective human population.
- Besides domestic pigs, swine HEV also infects wild boar.
- HEV infection commonly occurs at around 2–3 months of age as maternally derived antibodies decline.
- HEV seroprevalence is age-dependent, generally increasing after infection during the growing period.

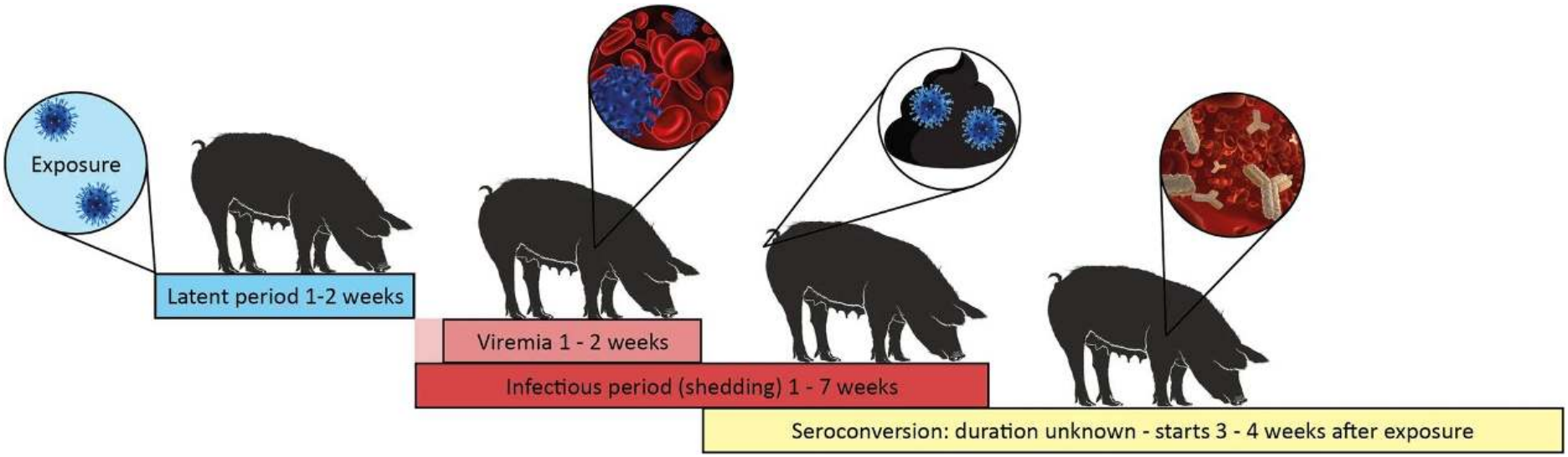
Transmission

- Swine HEV transmission among pigs is presumably fecal-oral, with feces from infected pigs serving as the primary source of the virus.
- Naïve pigs acquire infection through direct contact with infected pigs or through ingestion of feces-contaminated feed or water.

Pathogenesis

- HEV replicates in the gastrointestinal tract after ingestion and subsequently spreads to and replicates in its target organ, the liver.
- Extrahepatic sites of swine HEV replication have been identified in various tissues of pigs including small intestines, colon, and lymph nodes.

Course of HEV infection



Meester et al., 2021

- The HEV-infected pigs would continue to shed the virus in **feces** for **7 weeks**.

Clinical signs

- Pigs naturally and experimentally infected with swine HEV are asymptomatic.

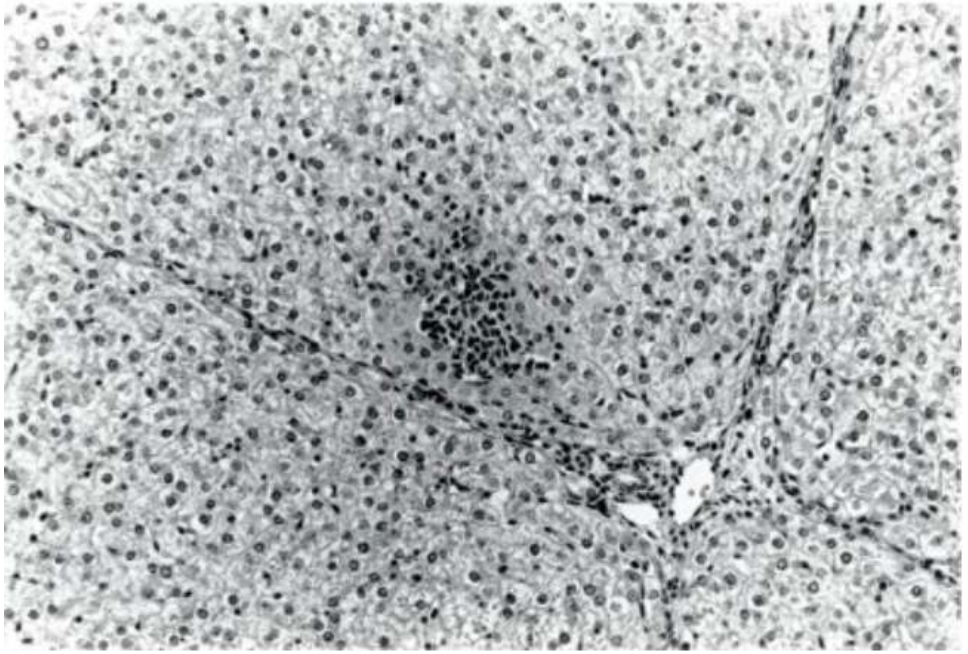
Gross lesions

- Pigs experimentally infected with swine HEV remained clinically normal, but mild-to-moderate enlargement of hepatic and mesenteric lymph nodes was observed.

Microscopic lesions

- Microscopic lesions characterized by mild-to-moderate multifocal lymphoplasmacytic hepatitis and focal hepatocellular necrosis are commonly observed in infected pigs.
- Hepatic inflammation and hepatocellular necrosis peaked in severity at 20 DPI.

Focal hepatocellular necrosis



Diseases of Swine, 11th Edition

Diagnostic samples

- **Serum or plasma** – anti-HEV antibodies and HEV RNA
- **Feces or rectal swabs** – HEV RNA detection by RT-PCR
- **Liver tissue** – HEV RNA detection, especially at necropsy or slaughter
- **Bile** – HEV RNA detection

Diagnosis – Laboratory

- Swine HEV cannot be efficiently propagated in cell culture, and only a few strains of swine HEV can be grown *in vitro*.
- Currently, the diagnosis of swine HEV infection is mainly based on PCR and ELISA.
- ELISA assays commonly use recombinant HEV capsid antigen, often derived from **HEV-3**.



Immunity

- The capsid protein of swine HEV is immunogenic and induces protective immunity.
- Piglets born to seropositive sows had maternal antibodies lasting 7-9 weeks, which confer protective immunity to the piglets against swine HEV infection.
- Prior infection of pigs with a **HEV-3** from swine protects against subsequent challenges with **HEV-3** and **HEV-4** strains from humans.

Prevention and Control

- The major concern for swine HEV is the zoonotic risk and its associated pork safety concerns.
- A commercial vaccine against HEV is currently licensed for **human** use only in China, but not elsewhere.
- Vaccination of pigs may help reduce zoonotic transmission and pork-associated HEV risk.
- Adequate personal and public hygiene can minimize swine HEV transmission on farms.



Take-home messages

- HEV infection is usually asymptomatic in pigs, but pigs are important reservoirs of zoonotic HEV.
- HEV-3 and HEV-4 can infect both pigs and humans and are important zoonotic genotypes.
- HEV spreads mainly by the fecal–oral route among pigs, while humans may be exposed through occupational contact or undercooked pork products.
- Control focuses on hygiene, reduction of environmental contamination, and thorough cooking rather than prevention of clinical disease in pigs.

Questions?

